



■ CS177 Bioinformatics: Software Development and Applications

Lecture 15: Transcriptomics

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Learning objectives



- At the end of this lecture, you will be able to:
 - Tell the meaning and measurement of gene expression
 - Explain how RNA-seq data are generated
 - Describe steps in a typical workflow of RNA-seq data analysis
 - Explain the motivation for single-cell RNA-seq (scRNA-seq) in regard to limitations of bulk RNA-seq
 - List challenges in the processing of scRNA-seq data
 - Describe how Autoencoders are used for scRNA-seq data processing and analysis
 - Describe the techniques and applications of spatial transcriptomics





Overview of Transcriptomics



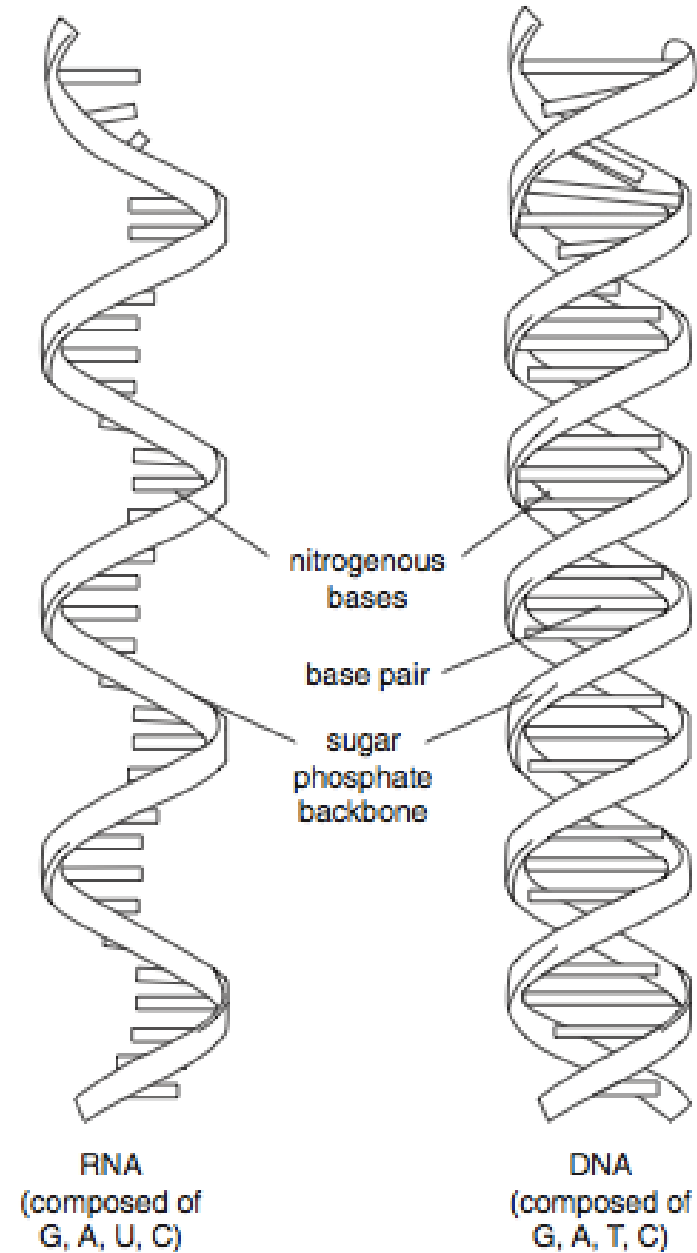


DNA and RNA



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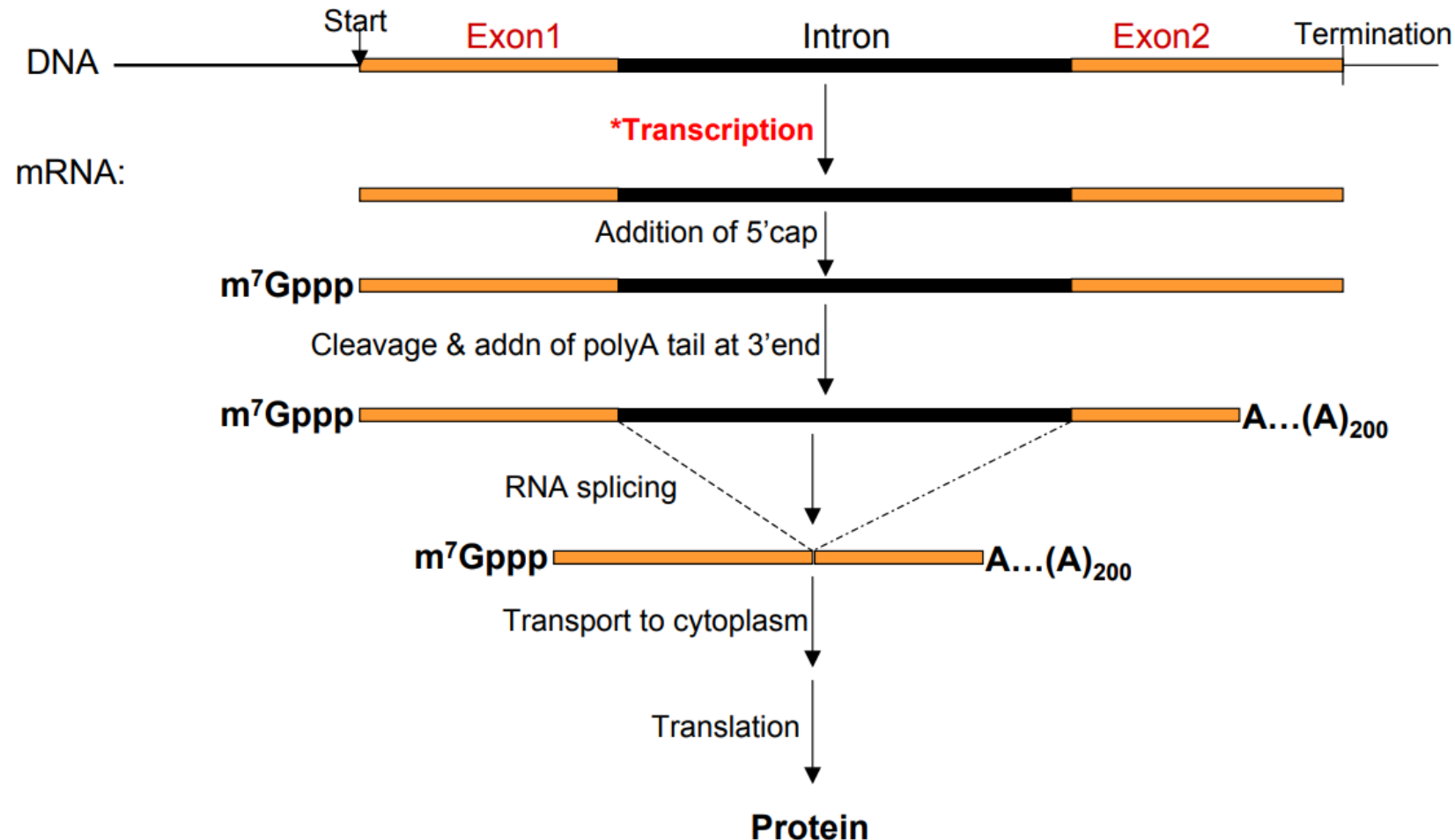
- In 1953, Crick and Watson discovered the double helical nature of DNA
- DNA: A, G, C, T (adenine, guanine, cytosine, thymidine)
- RNA: G, A, U, C (U = uracil)
- Crick further helped elucidate the nature of the genetic code, proposing the existence of tRNA



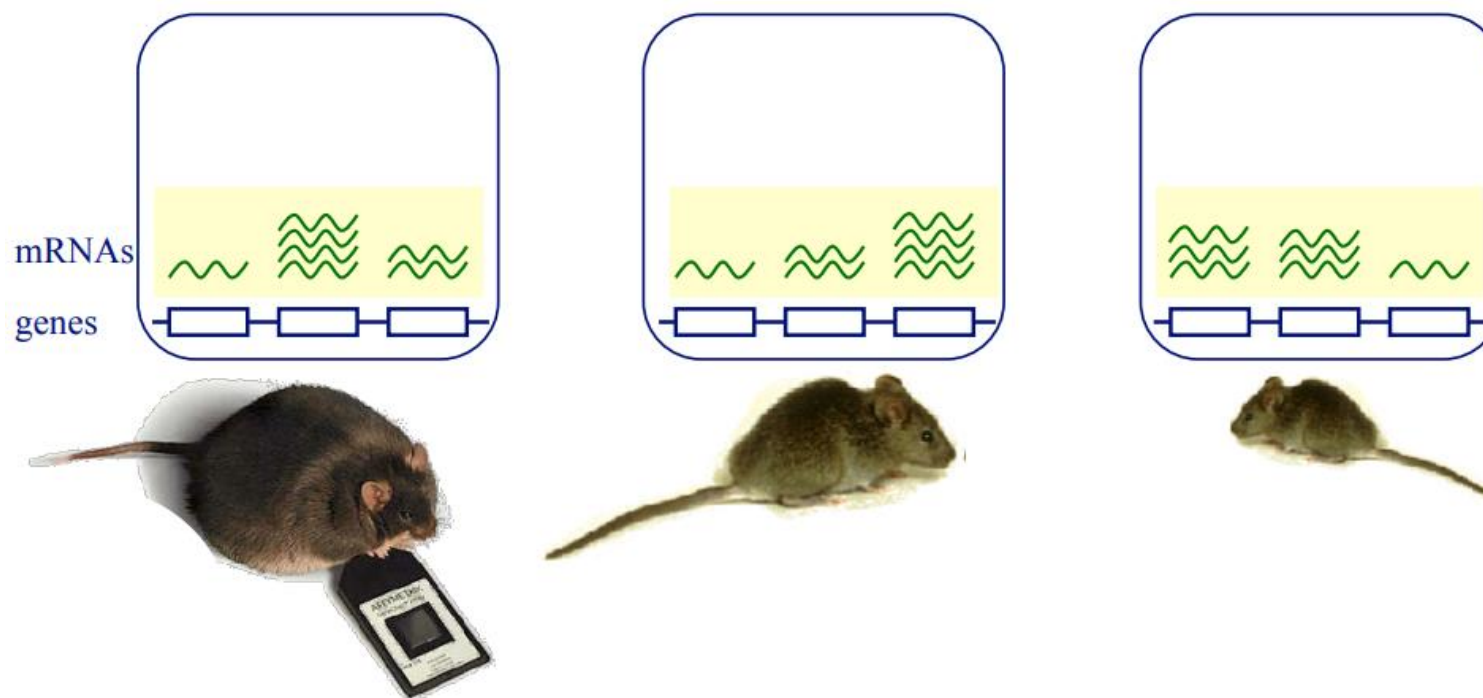
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Regulation of gene expression

- Any of these stages could be used to regulate expression of specific genes in particular tissues
- But in general, the primary control of gene expression is at the level of transcription



High-throughput data of RNA abundances



- A common high-throughput study: measuring RNA abundances
- **What is varied:** individuals, strains, cell types, environmental conditions, disease states, etc.
- **What is measured:** RNA quantities for thousands of genes, exons or other transcribed sequences



Low and high-throughput technologies to study mRNAs



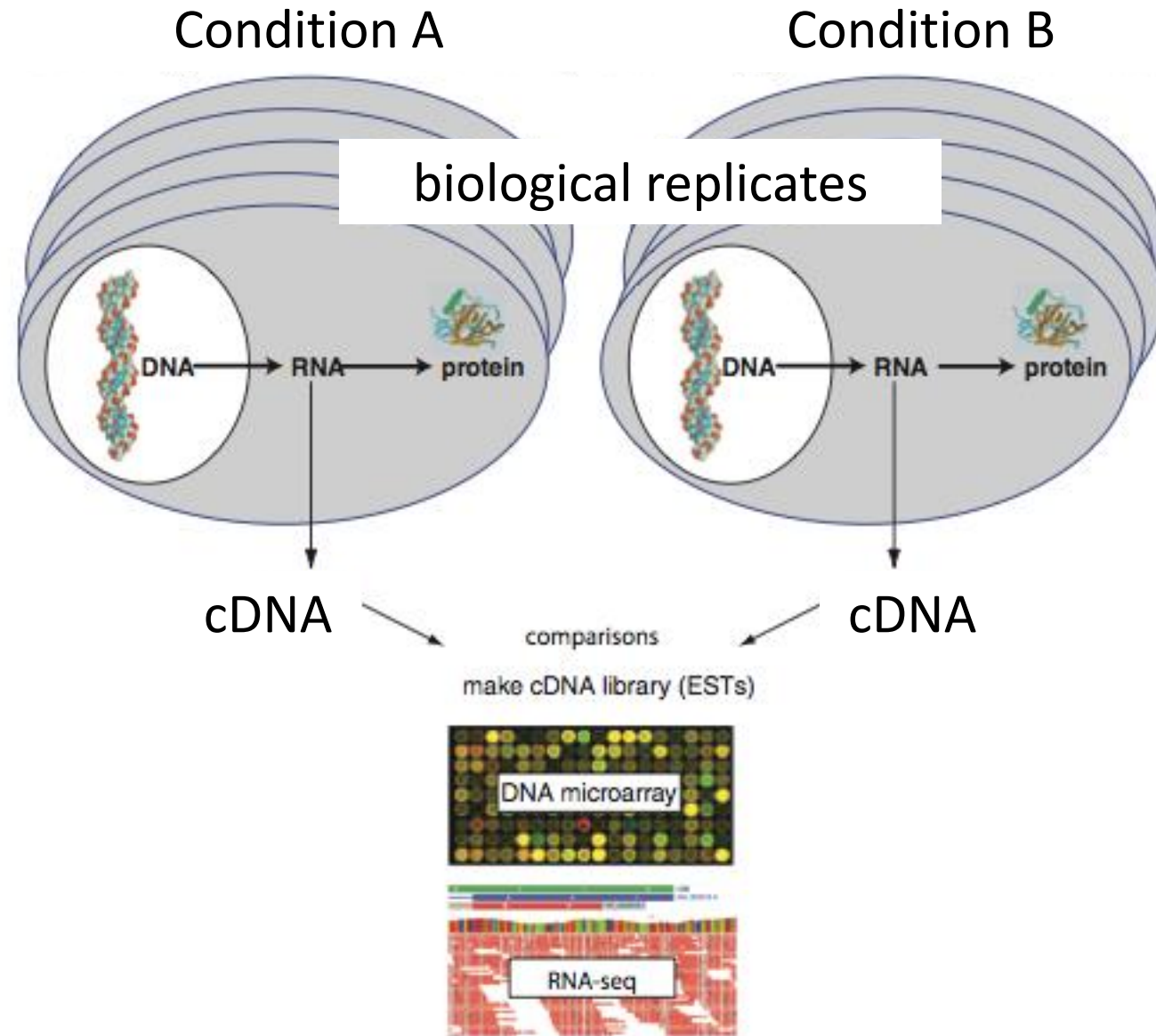
- Three techniques for the study of mRNA:
 - complementary DNA (cDNA) libraries
 - microarrays (e.g. using the Affymetrix platform)
 - RNA-seq

Low throughput techniques (e.g. Northern blots, PCR) may seem laborious and able to provide only limited amounts of information.

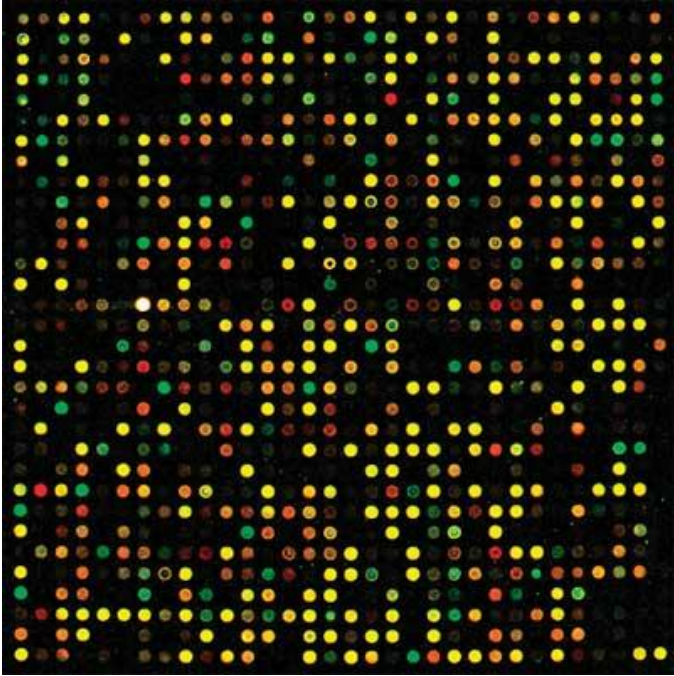
Yet they also serve as the trusted “gold standards” and provide crucial validation of high throughput techniques.



Gene expression measured with high-throughput technologies



Measuring transcription the old way: microarrays



- Each spot has “probes” for a certain gene
- **Probe**: a DNA sequence complementary to a certain gene
- Relies on **complementary hybridization**
- Intensity/color of light from each spot is measurement of the number of transcripts for a certain gene in a sample
- Requires knowledge of gene sequences





RNA-seq

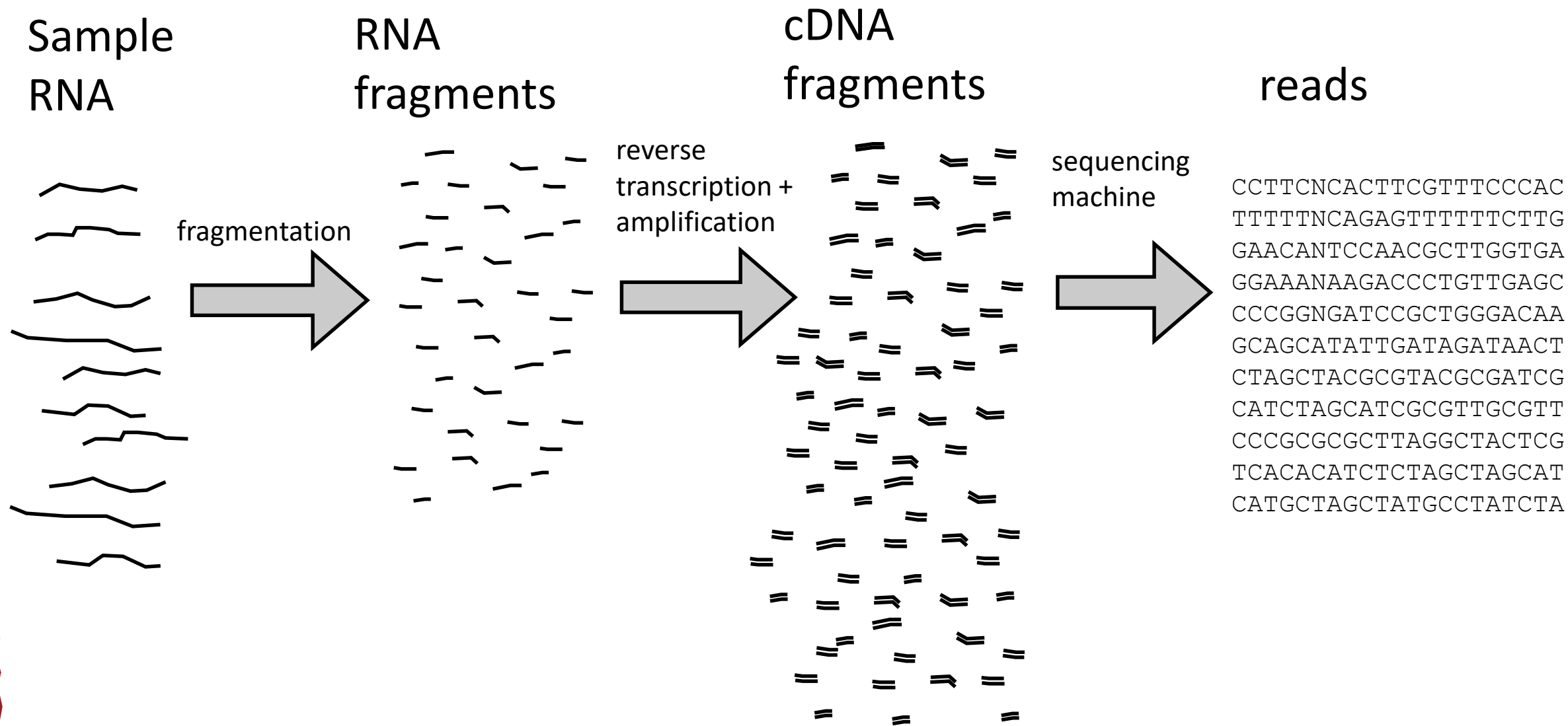


- RNA-seq is used to accomplish the same goal as microarrays: quantifying RNA transcript levels. However, it was considered revolutionary because
 - It allows the measurement of essentially **all** RNA transcripts (rather than only those pre-selected on a microarray surface)
 - It has a broader dynamic range
 - It allows identification of **novel** transcripts and transcript isoforms, and
 - It is able to quantify **alternative splicing** events

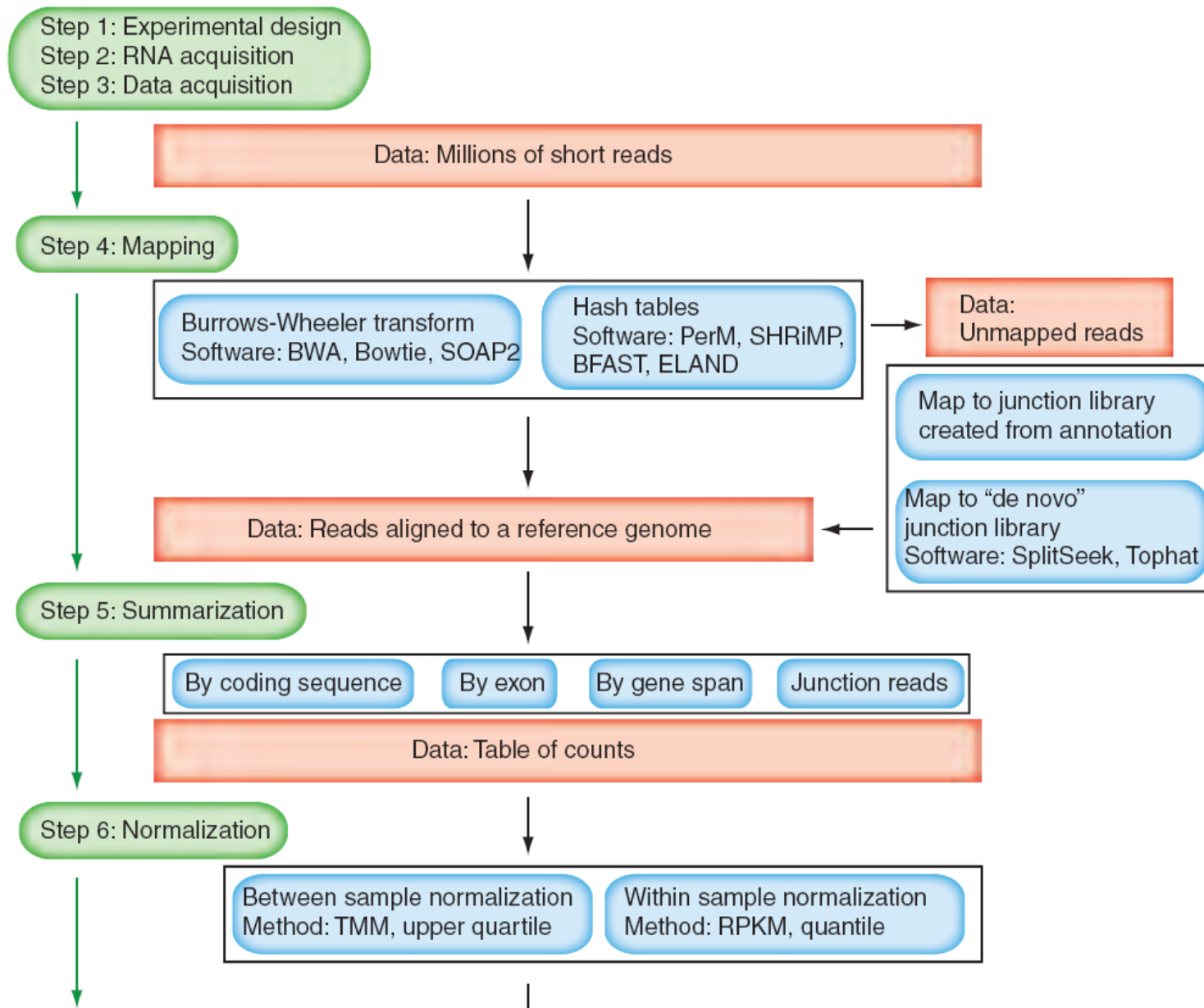




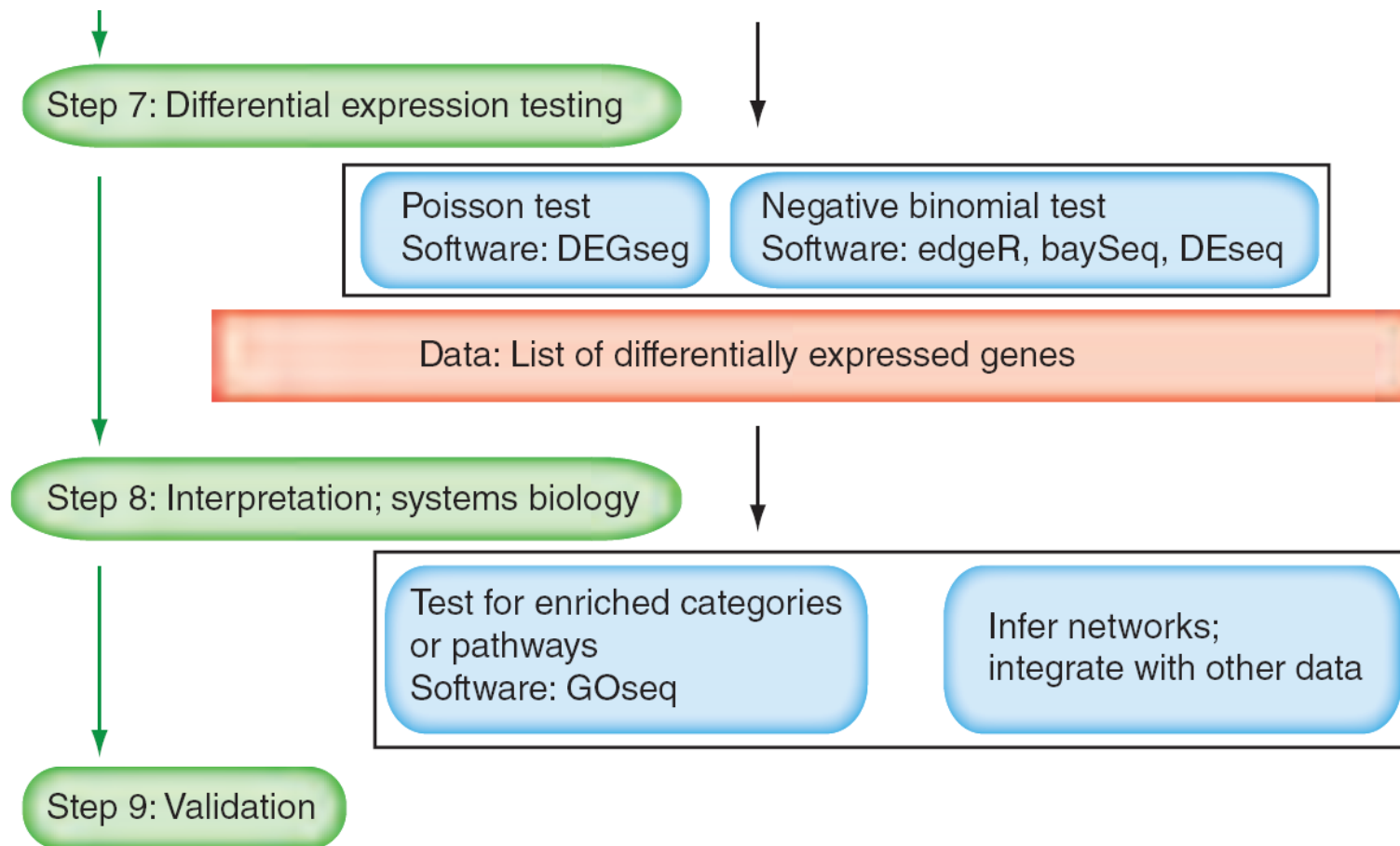
A generic RNA-seq protocol



Workflow for RNA-seq data analysis (1/2)



Workflow for RNA-seq data analysis (2/2)





■ Tasks with RNA-seq data analysis

- **Assembly:**
 - Given: RNA-seq reads (and possibly a genome sequence)
 - Do: Reconstruct full-length transcript sequences from the reads
- **Quantification (**our focus**):**
 - Given: RNA-seq reads and transcript sequences
 - Do: Estimate the relative abundances of transcripts (“gene expression level”)
- **Differential expression:**
 - Given: RNA-seq reads from two different samples and transcript sequences
 - Do: Predict which transcripts have different abundances between two samples





Basic steps of quantification algorithms

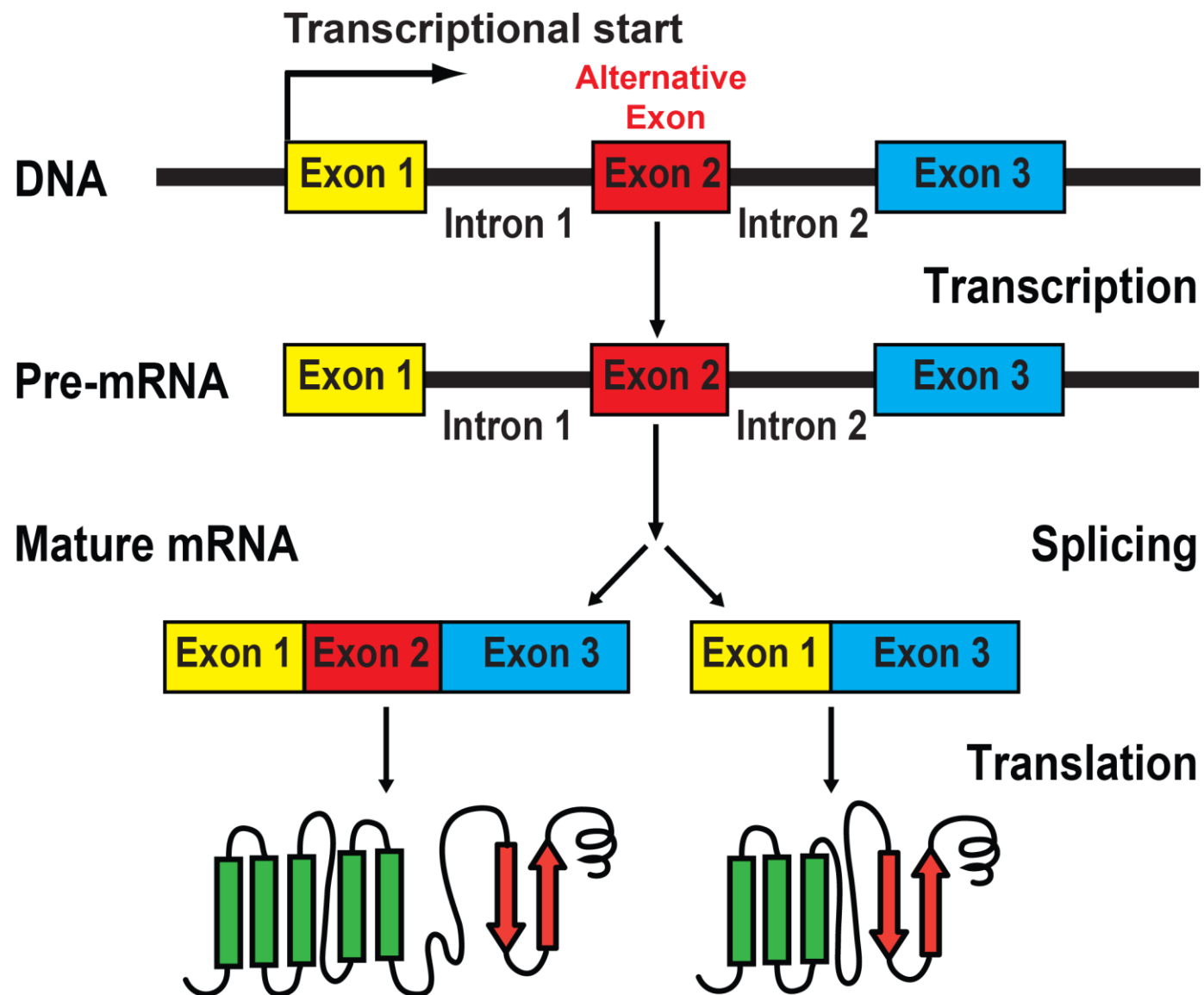


- Align reads against a set of reference transcript sequences
- Count the number of reads aligning to each transcript
- Convert read counts into relative expression levels





Alternative splicing





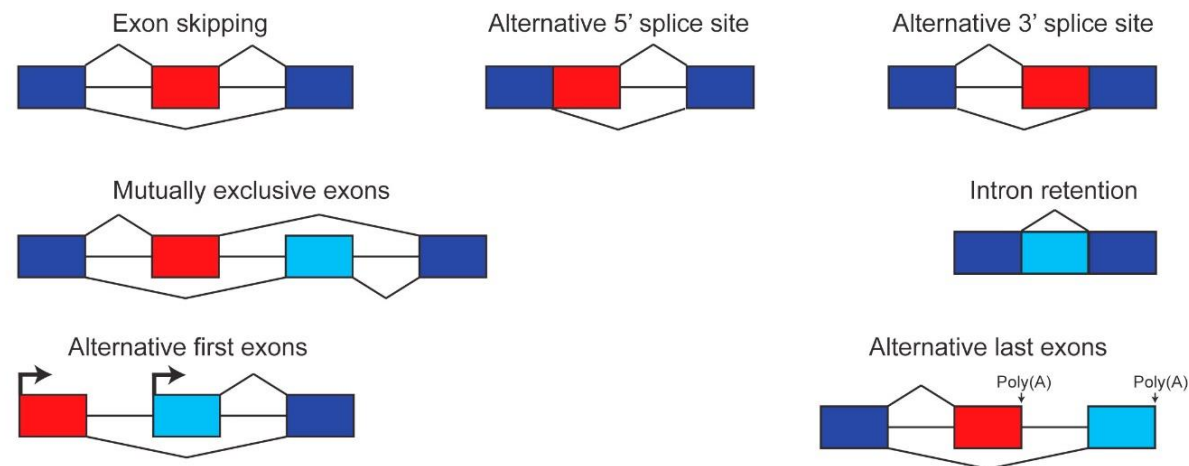
Prevalence and importance of alternative splicing



- Almost all human genes undergo alternative splicing to produce mRNA and protein isoforms
- Important in regulation of gene function
- Aberrant splicing is a major cause of human diseases

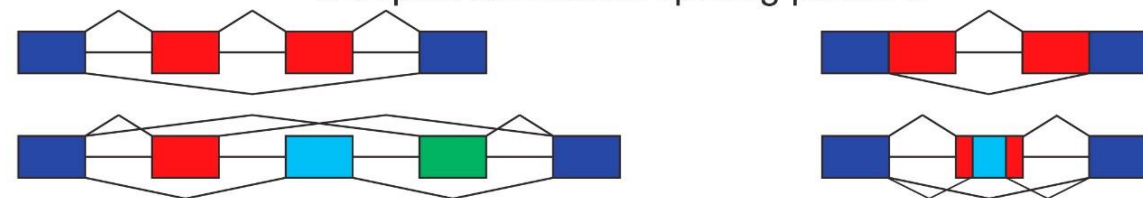
A

Basic alternative splicing patterns



B

Complex alternative splicing patterns



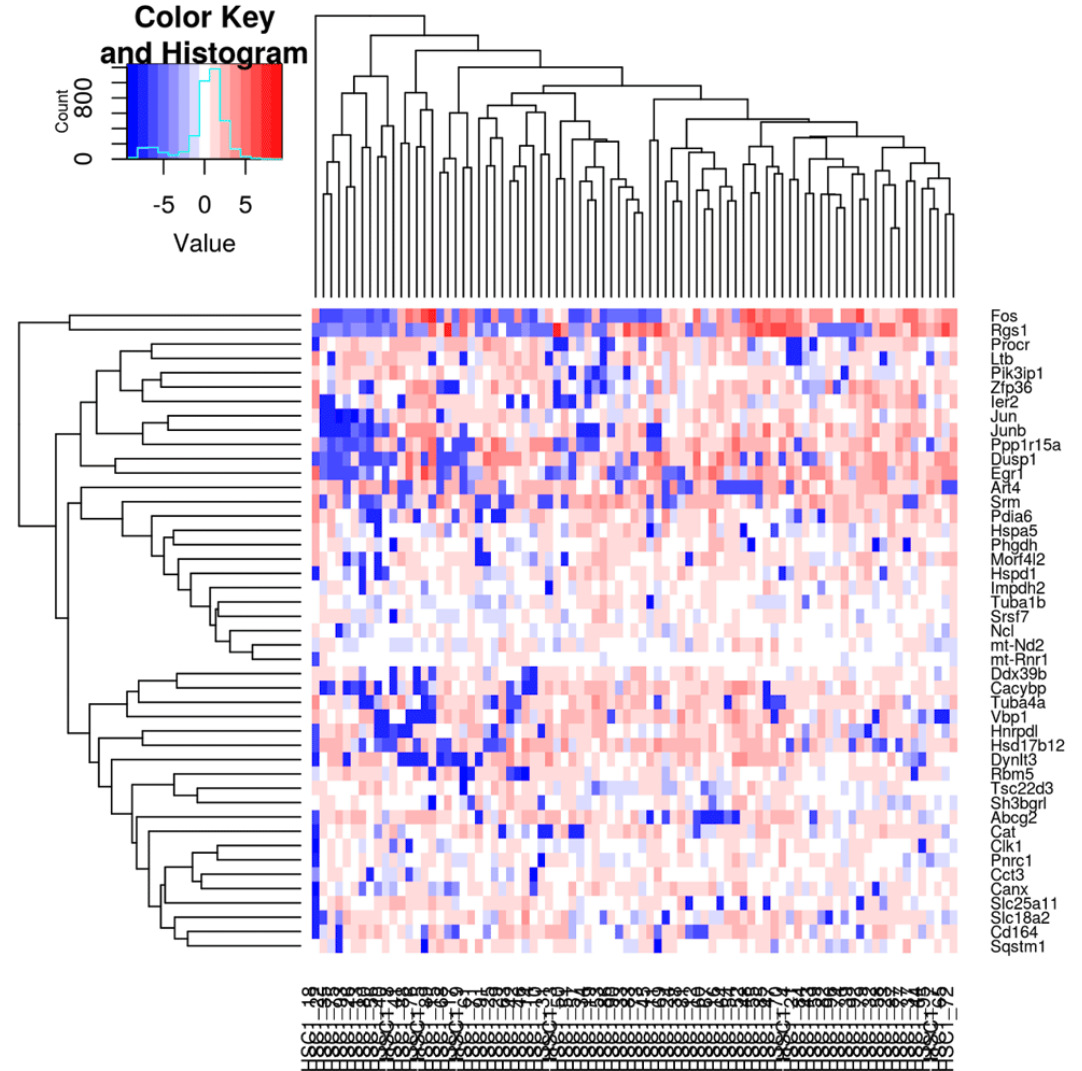
Park et al., *Am. J. Hum. Genet.*, 2018



Downstream analyses



- Principal component analysis (PCA)
 - Reduce dimensionality
 - Feature selection
 - Identify groups of objects (e.g. genes/samples)
 - Visualization
 - Identify outliers
- Identify differentially expressed genes
- Clustering analysis
- Classification



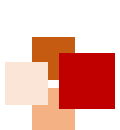
Dendrograms formed by hierarchical clustering on the Euclidean distances between genes (row) or samples (column)



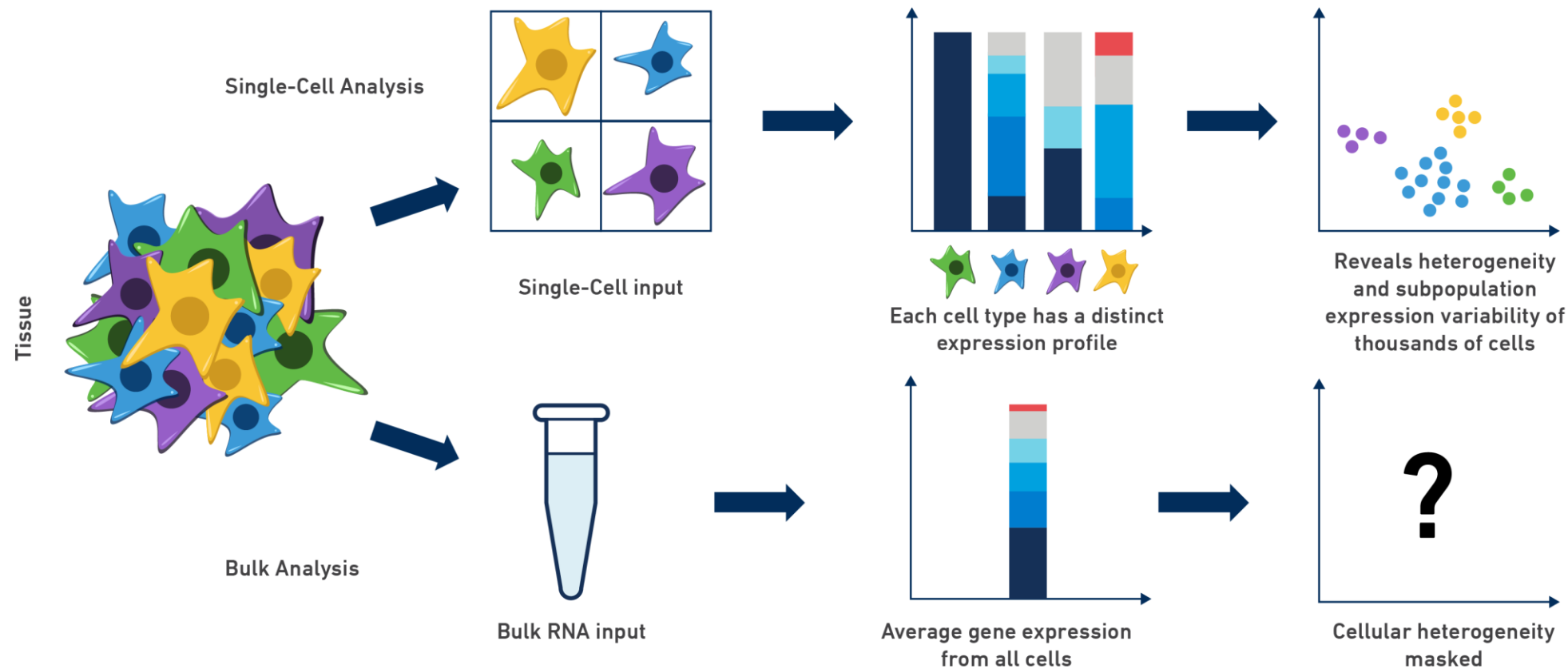
Single-cell RNA Sequencing (scRNA-seq)

Some PPT slides of this module are courtesy from Dr. Lichun Jiang, SIAIS, ShanghaiTech





Issue of heterogeneity in bulk RNA-seq

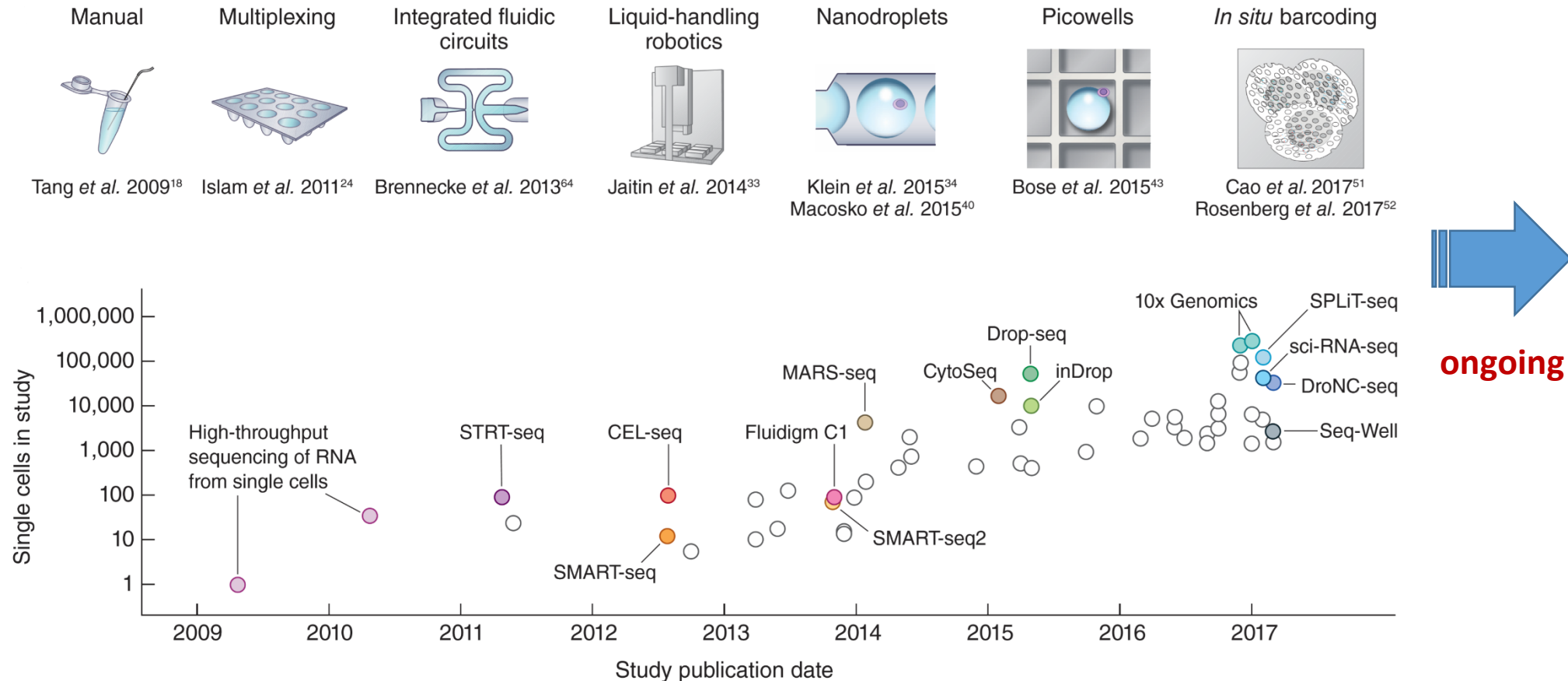


<https://media.biocompare.com/m/37/article/o/345311fig1.png>



Scales of scRNA-seq experiments

- Development of new methods and protocols for scRNA-seq is an active area of research
- Key technologies allowed jumps in scales during the past 10 years

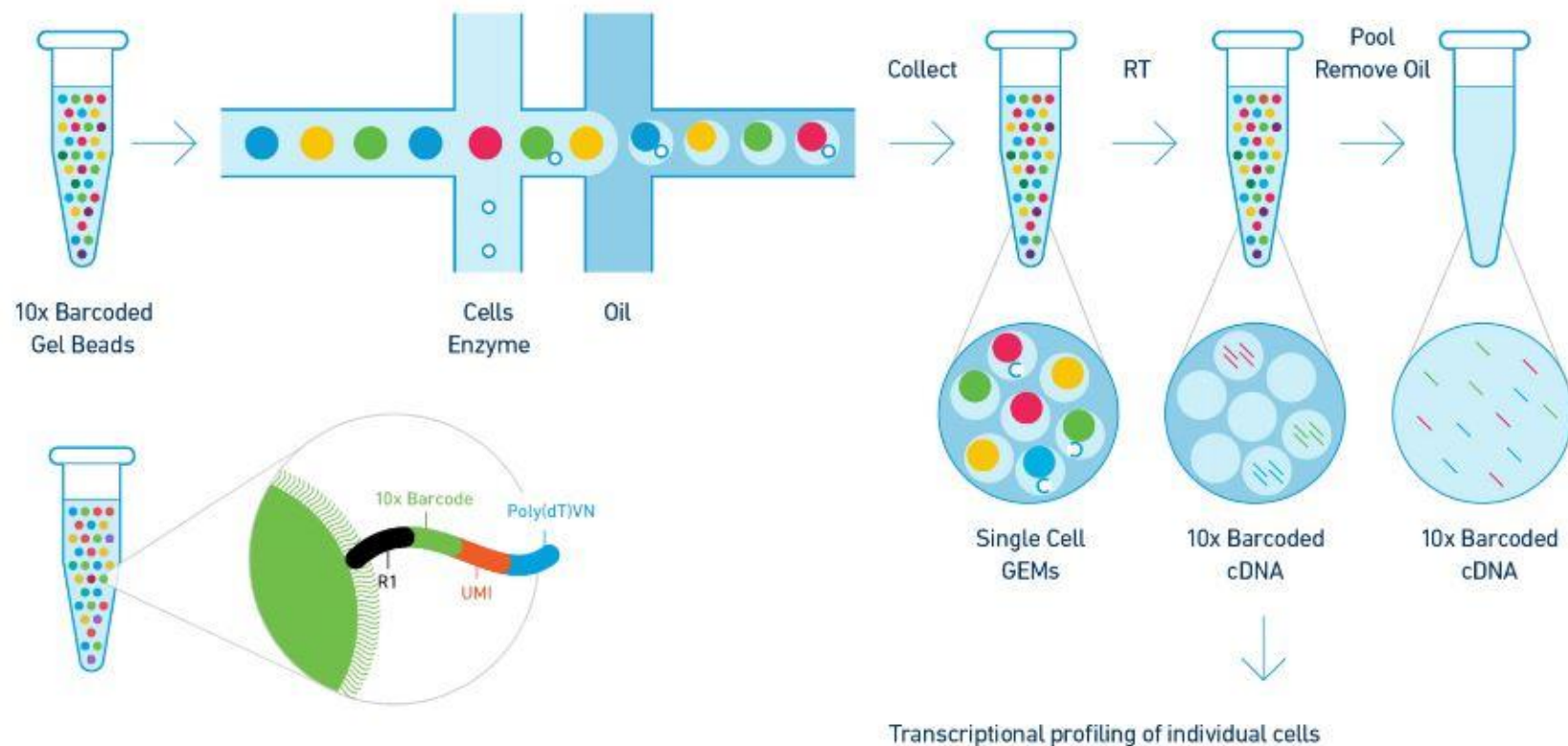
Svensson *et al.*, Nature Protocols, 2018



Example of scRNA-seq experiment

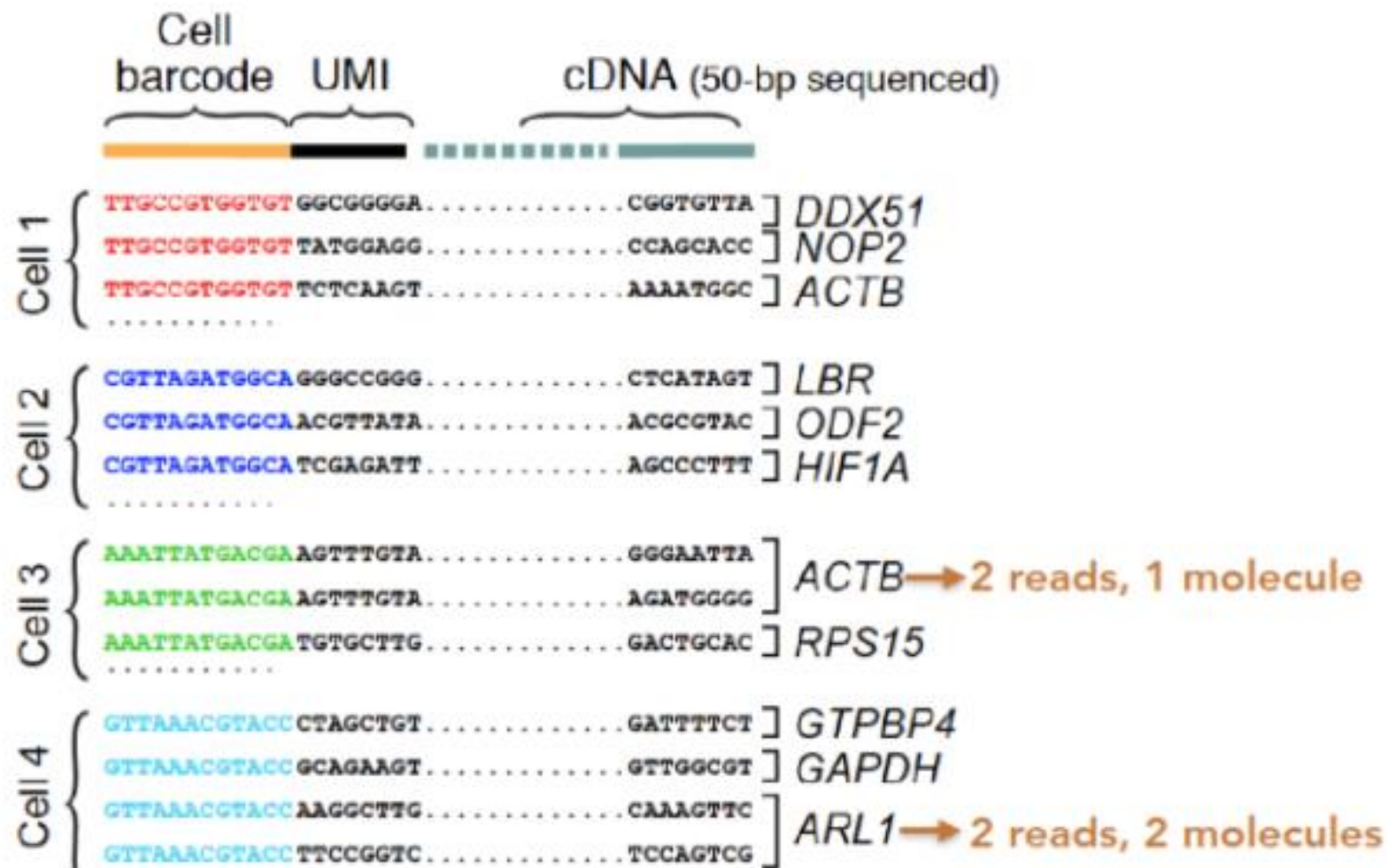


<https://www.10xgenomics.com/products/single-cell-gene-expression/>



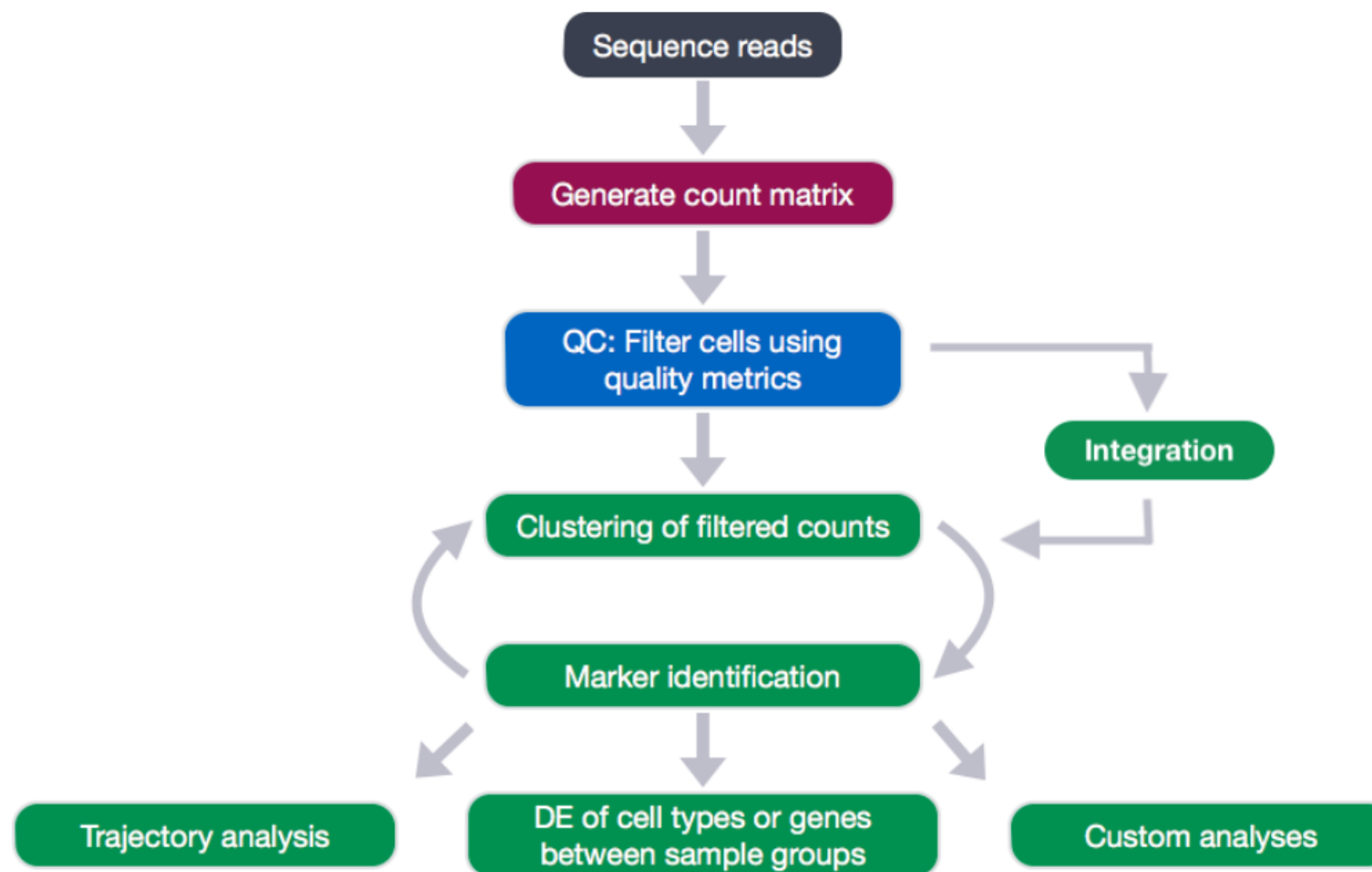


Toy data example





Workflow of scRNA-seq data analysis



https://hbctraining.github.io/scRNA-seq/lessons/04_SC_quality_control.html





Count matrix of scRNA-seq data



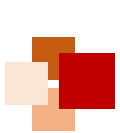
Toy datasets

G/C	C1	C2	C3	...	Cn
G1	2	5	0	...	8
G2	0	19	40	...	0
G3	230	0	398	...	0
...	...	...	...	...	...
Gm	0	0	0		3000

In a single piece of data:

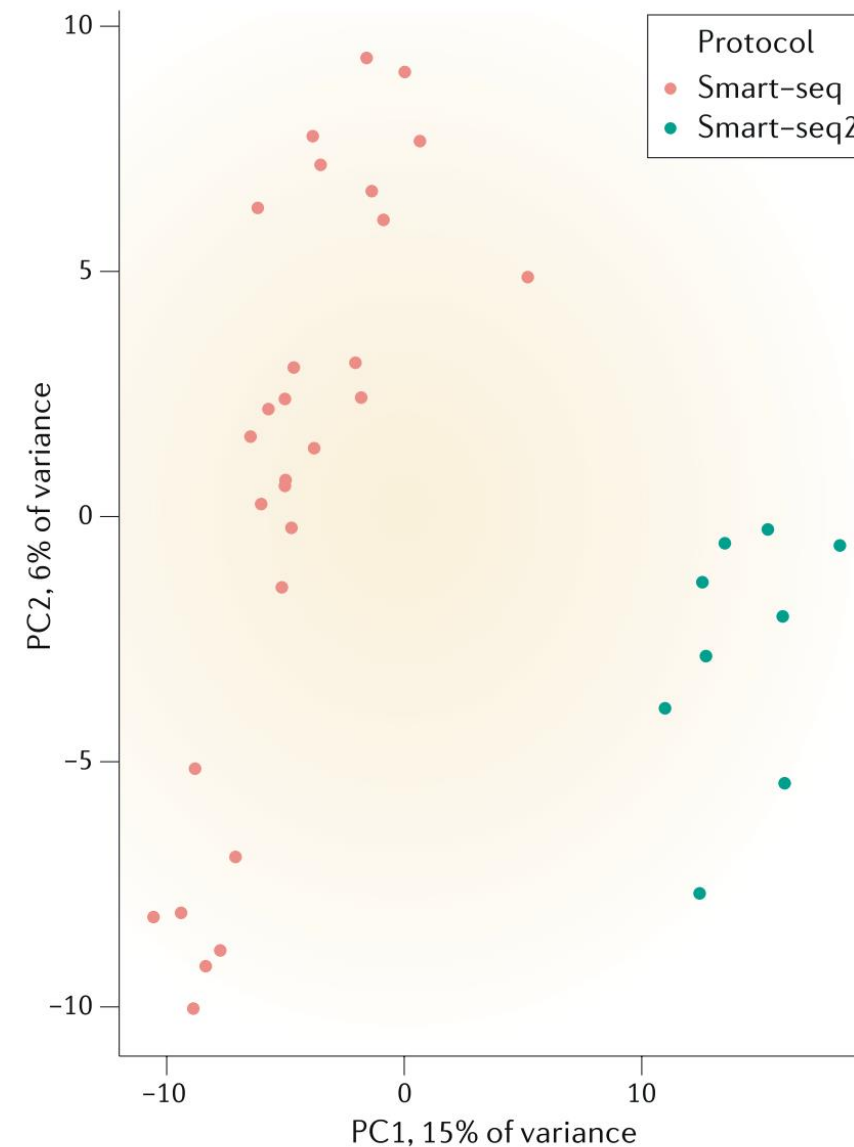
- m in the scale of 10K, with dropout
- n in the scale of 10K or more, dynamic range
- The n will increase with more data and studies, and technique improvement





Batch effect

- Batch removal not necessarily helps
- Need careful interactive sanity check



Kiselev et al., Nature Reviews, 2019



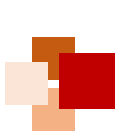


Dropout events



- **Spurious zero entries** due to
 - Small numbers of RNA molecules in single cell
 - Stochastic nature of transcriptional process
- Such spurious zero expression values are called **dropout events**
 - Accidentally, the term of “dropout” also refers to the regularization technique to reduce overfitting in deep learning

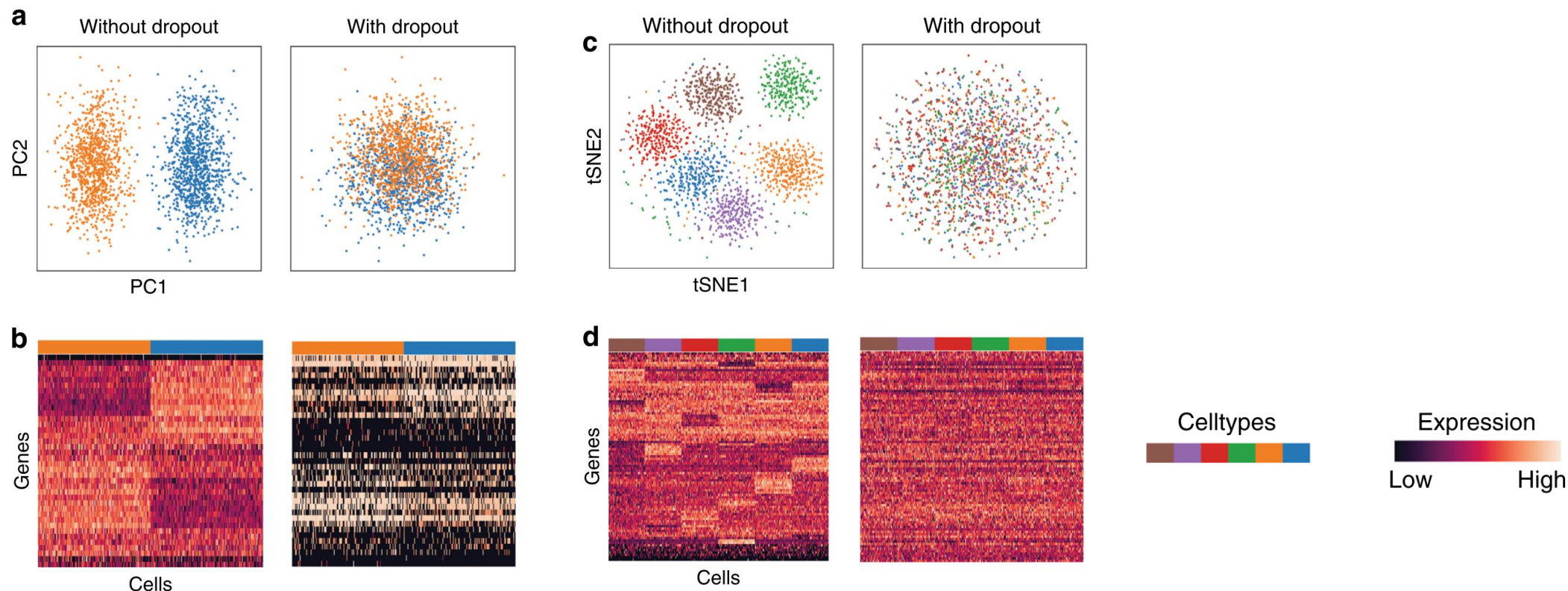




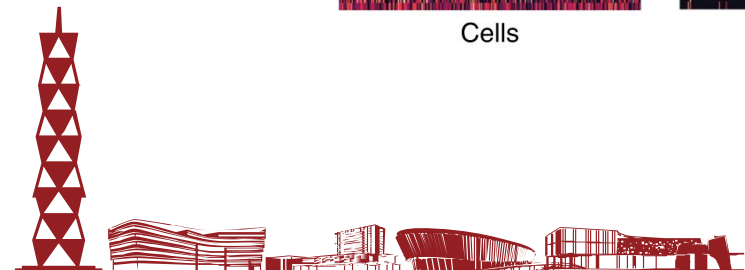
Dropout events (continued)



- Dropouts due to technical limitations of scRNA-seq can confound downstream analysis

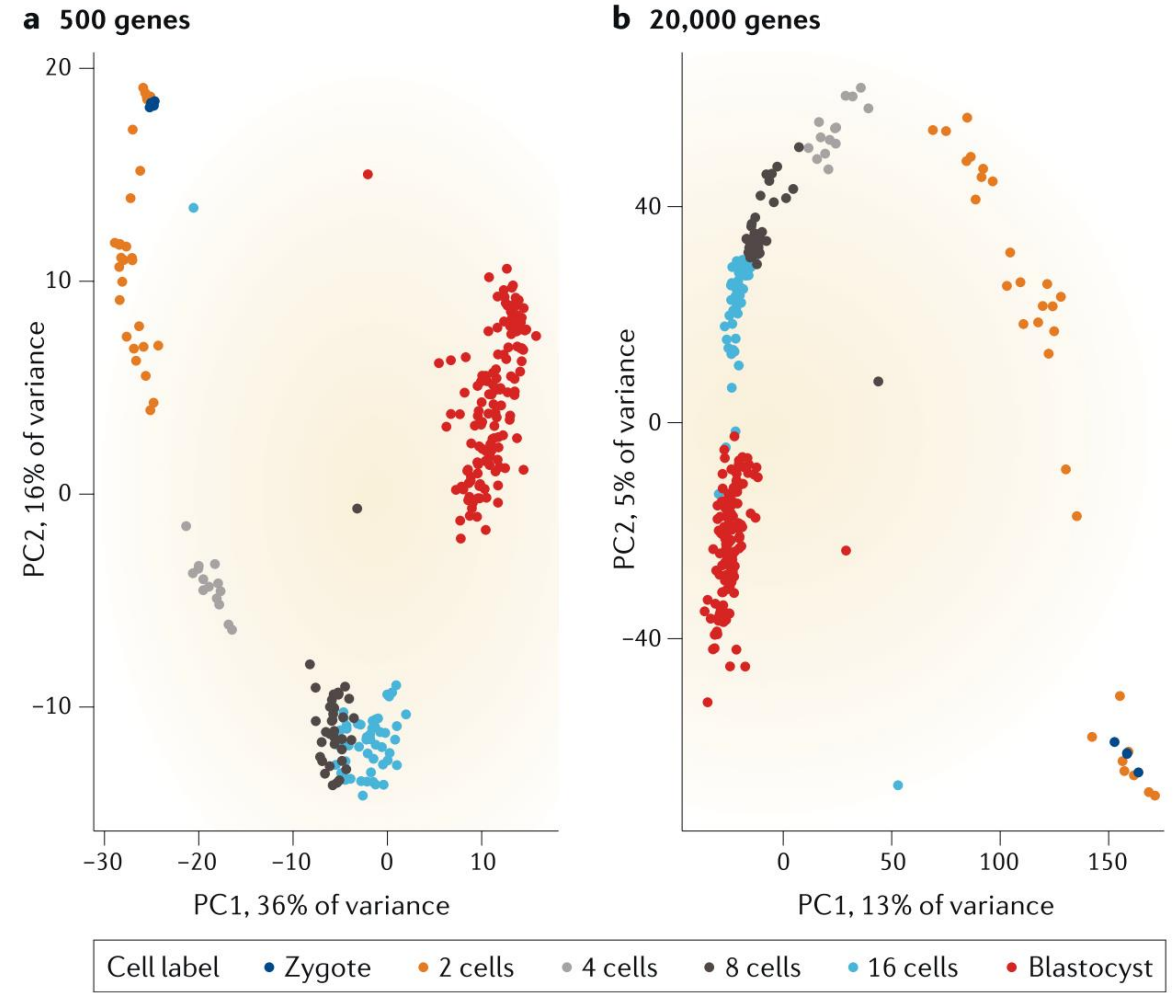


Eraslan et al., Nature Communications, 2019

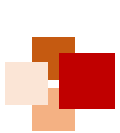


Curse of dimensionality

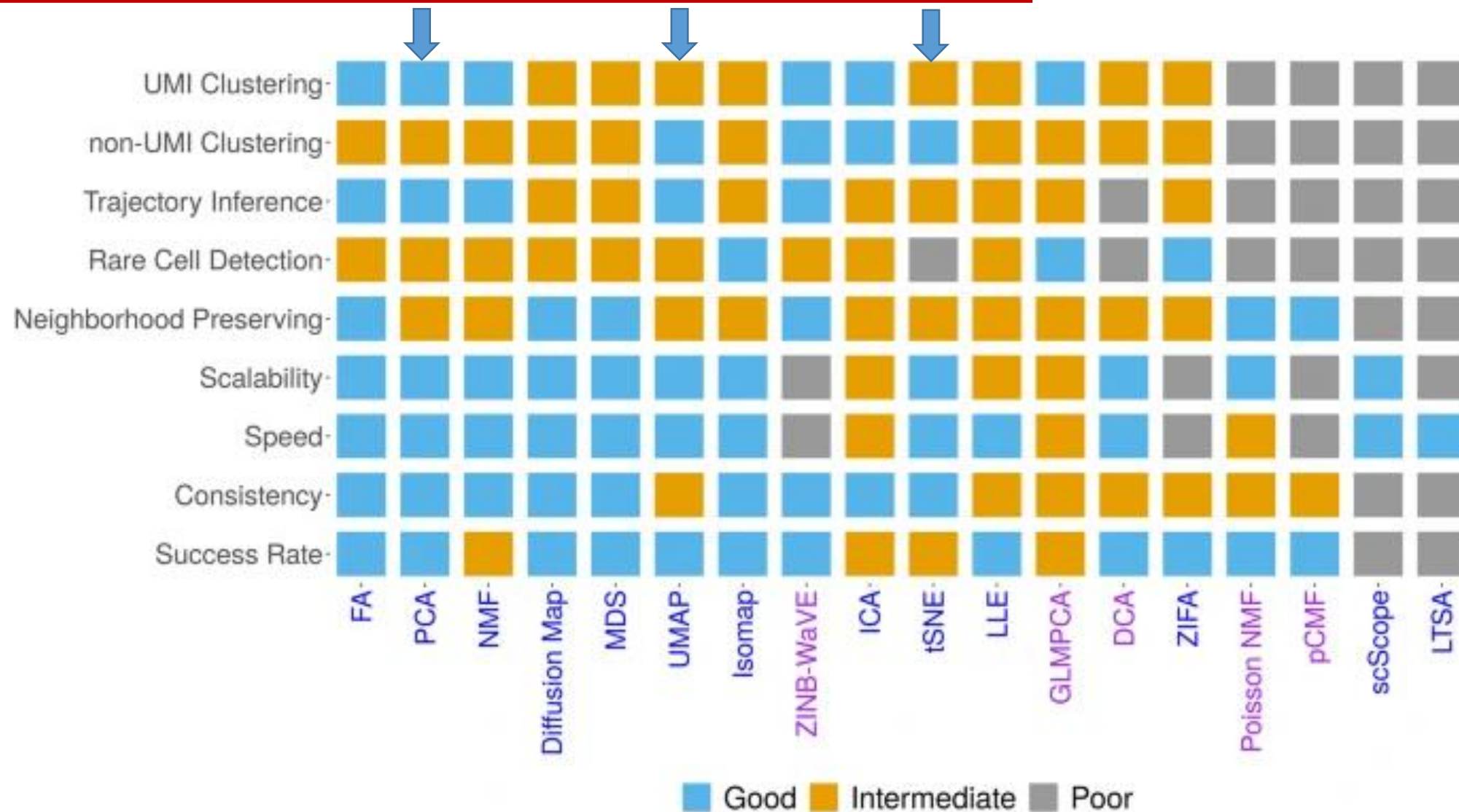
- In a high-dimensional space, as the data points become sparse, the distance measures (e.g. Euclidean distance) lose their effectiveness, rendering the concept of nearest neighbor unclear and data analysis problems difficult. This is called the **curse of dimensionality**
- One way to alleviate the issue caused by high dimensionality is **to increase the data**
 - Technique limitation on increasing data
- A more feasible solution is to do **dimensionality reduction**



Kiselev et al., Nature Reviews, 2019

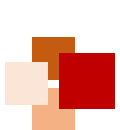


Dimension reduction methods



Sun et al. *Genome Biology* **20**, 269 (2019)

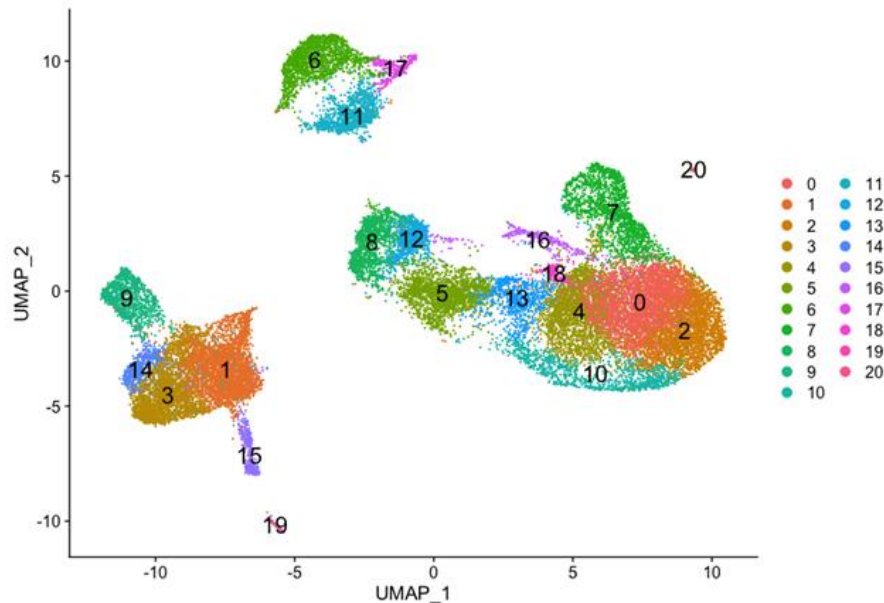
<https://genomebiology.biomedcentral.com/articles/10.1186/s13059-019-1898-6>



Clustering and cell identity assignment



- Marker gene analysis for each cluster
- Interactively assign identity based on prior knowledge



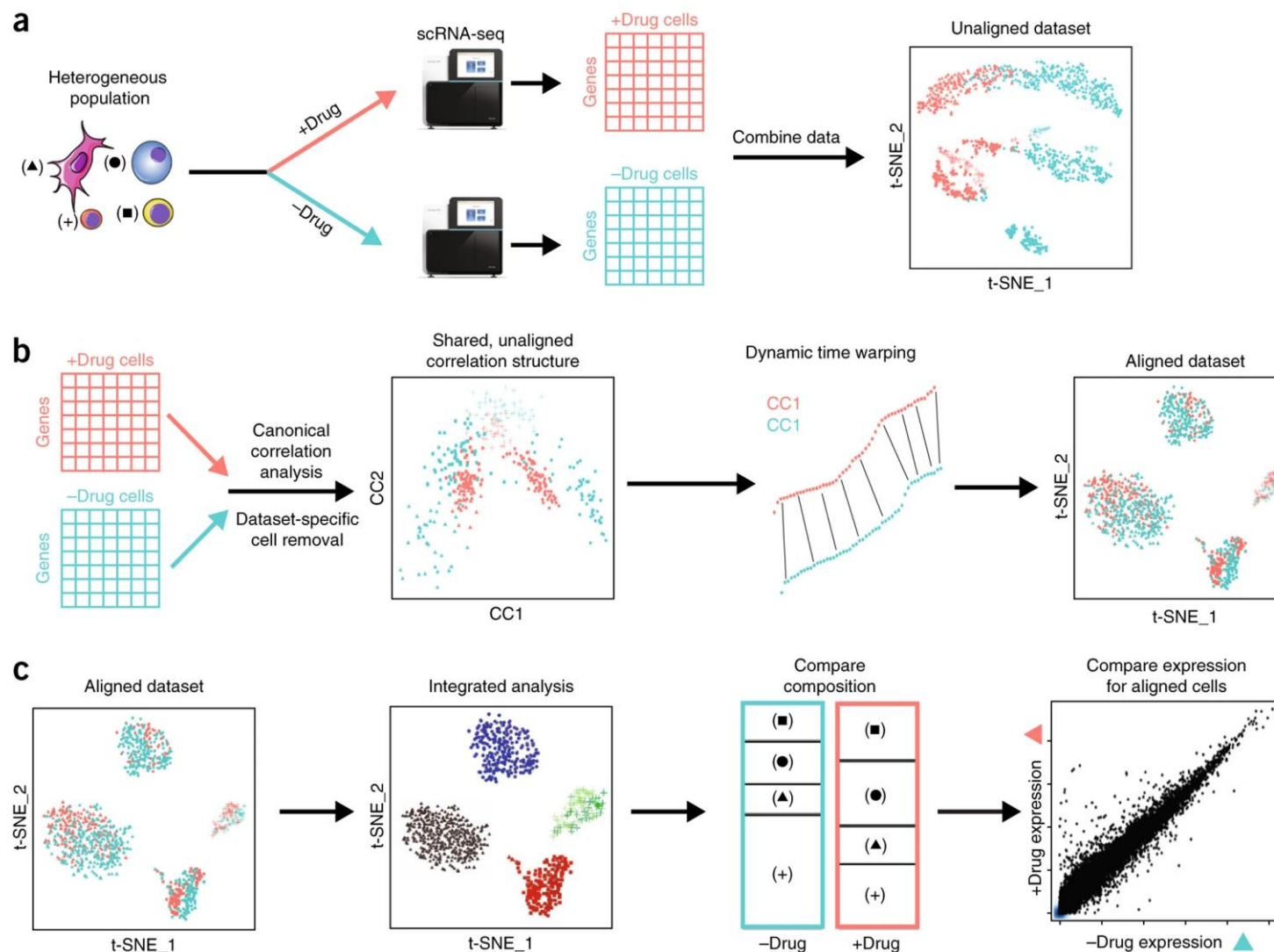
Marker gene:

- Average expression difference between clusters
- Percentage of cells in clusters with positive expression

K-nearest neighbor (KNN) graph



One of the popular tools -- Seurat



Butler et al. Nature Biotech. vol. 36, pp. 411–420 (2018)

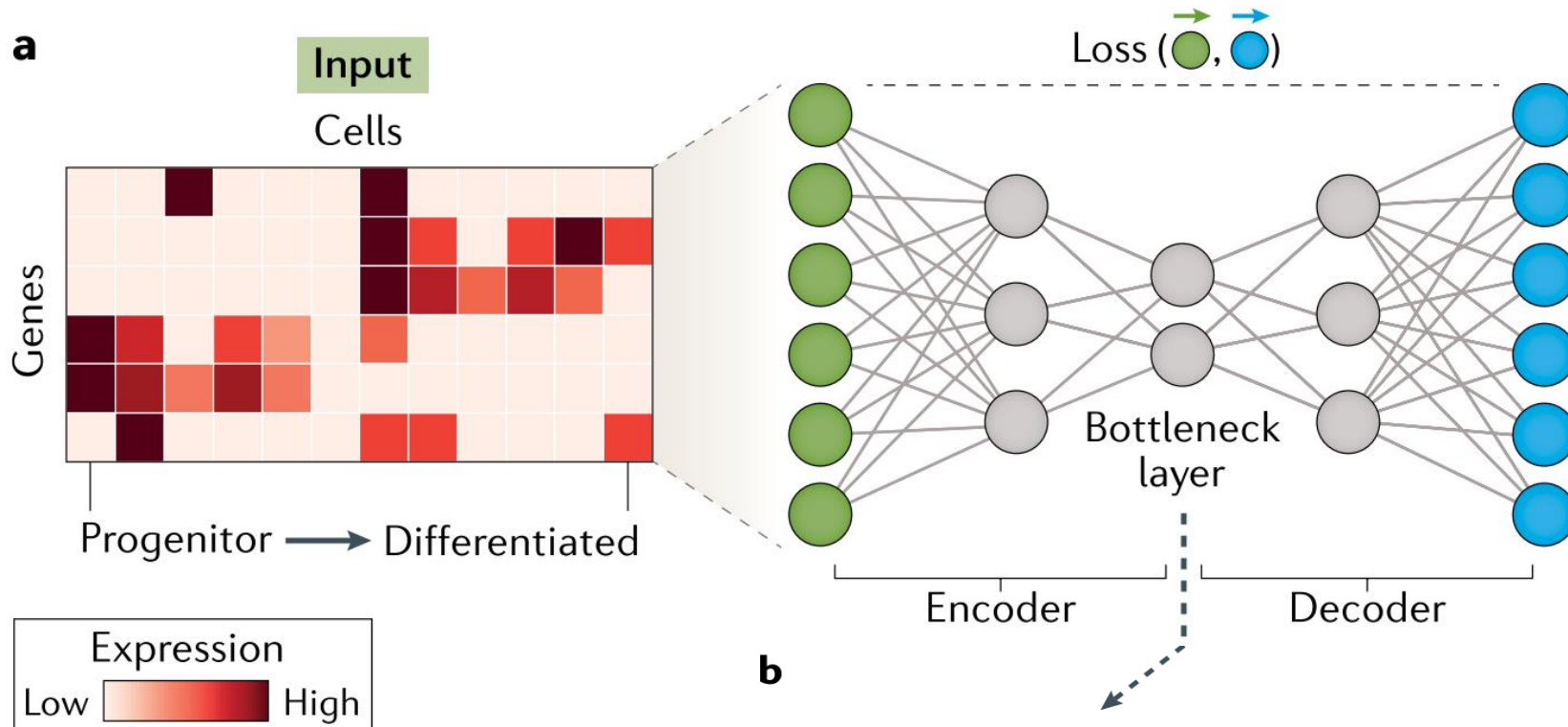
Hao et al. Cell, vol. 184, pp. 3573–3587 (2021)



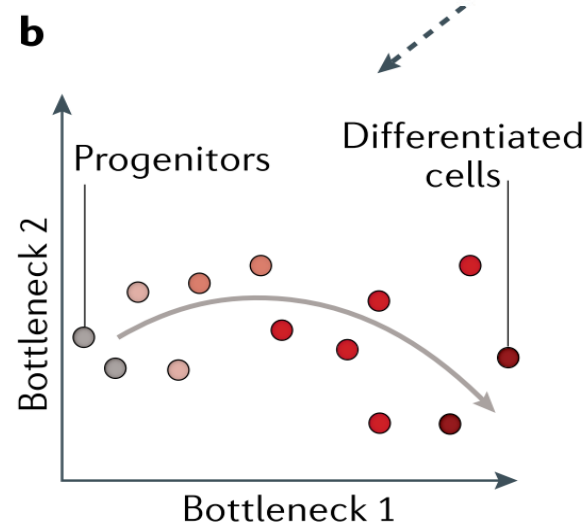
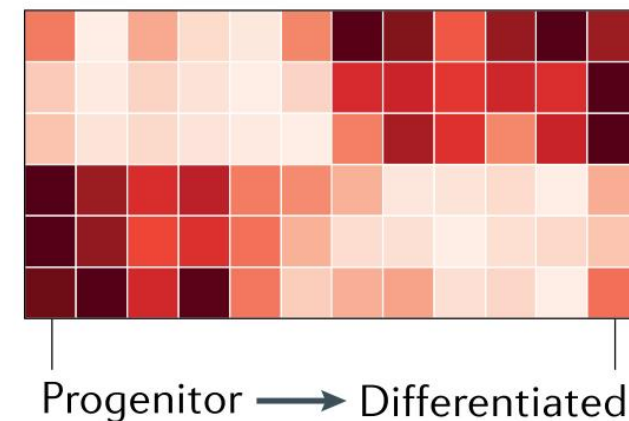
Autoencoders for scRNA-seq data



Recap: Autoencoder



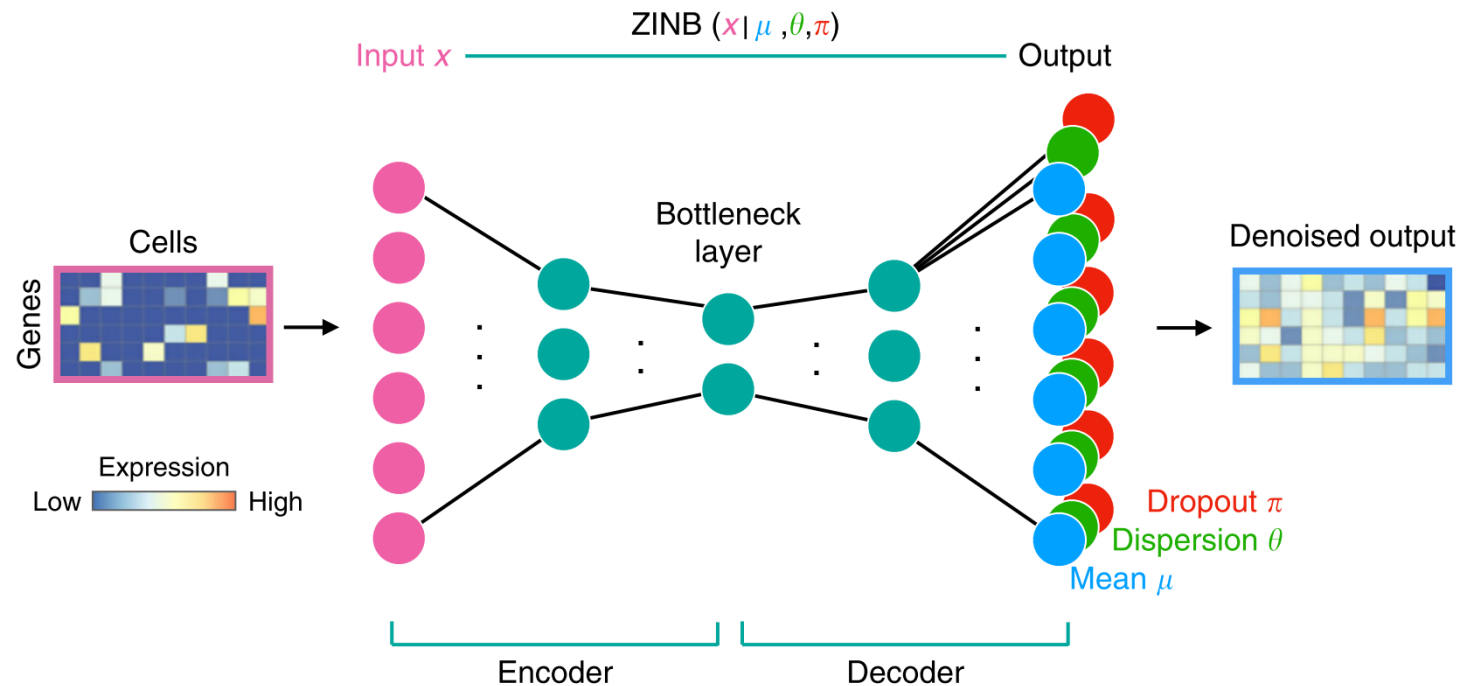
Reconstruction



Eraslan et al., Nature Reviews Genetics 20, 389-403 (2019)

DCA (deep count autoencoder)

- **DCA** is an autoencoder-based method for denoising scRNA-seq datasets
- It has a **ZINB** (zero-inflated negative binomial) loss function
 - **Blue nodes** depict the mean of the negative binomial distribution which is the main output of the method representing denoised data, denoted as μ
 - **Green nodes** represent the dispersion of the ZINB distribution, denoted as θ
 - **Red nodes** represent the dropout of the ZINB distribution, denoted as π



Eraslan et al., Nature Communications, 2019



DCA noise model



- The ZINB distribution models highly sparse and overdispersed count data
- The ZINB mixture model consists of two components:
 - a **point mass at zero** which represents excess zero values in the data
 - a **negative binomial component** representing the count distribution
- The **ZINB distribution** is as below:

$$\text{NB}(x; \mu, \theta) = \frac{\Gamma(x + \theta)}{\Gamma(\theta)} \left(\frac{\theta}{\theta + \mu}\right)^\theta \left(\frac{\mu}{\theta + \mu}\right)^x$$

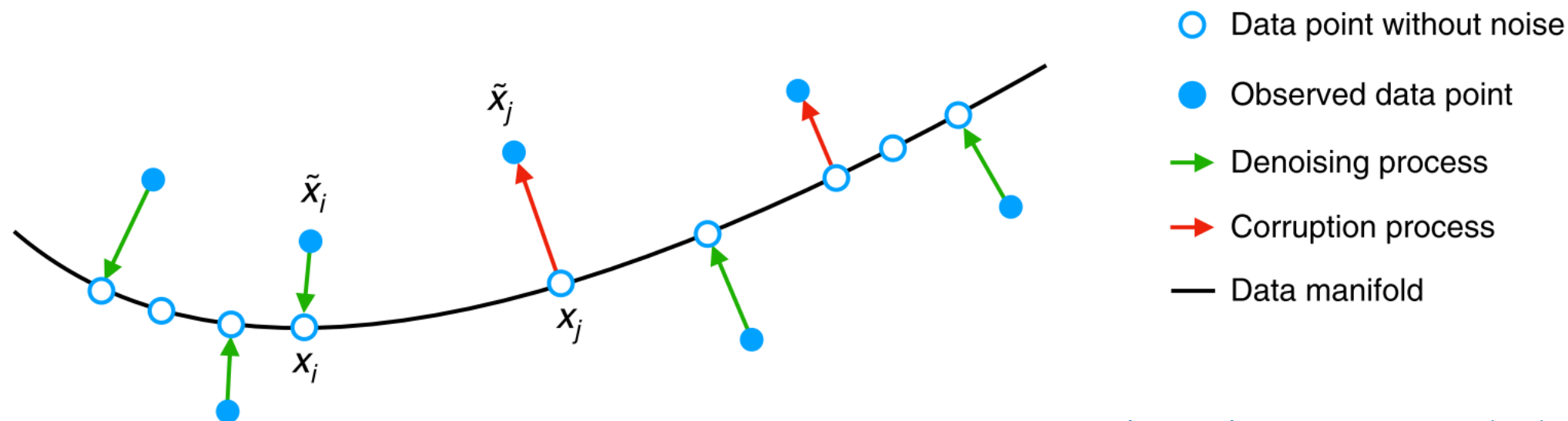
$$\text{ZINB}(x; \pi, \mu, \theta) = \pi \delta_0(x) + (1 - \pi) \text{NB}(x; \mu, \theta)$$



Geometric view of DCA

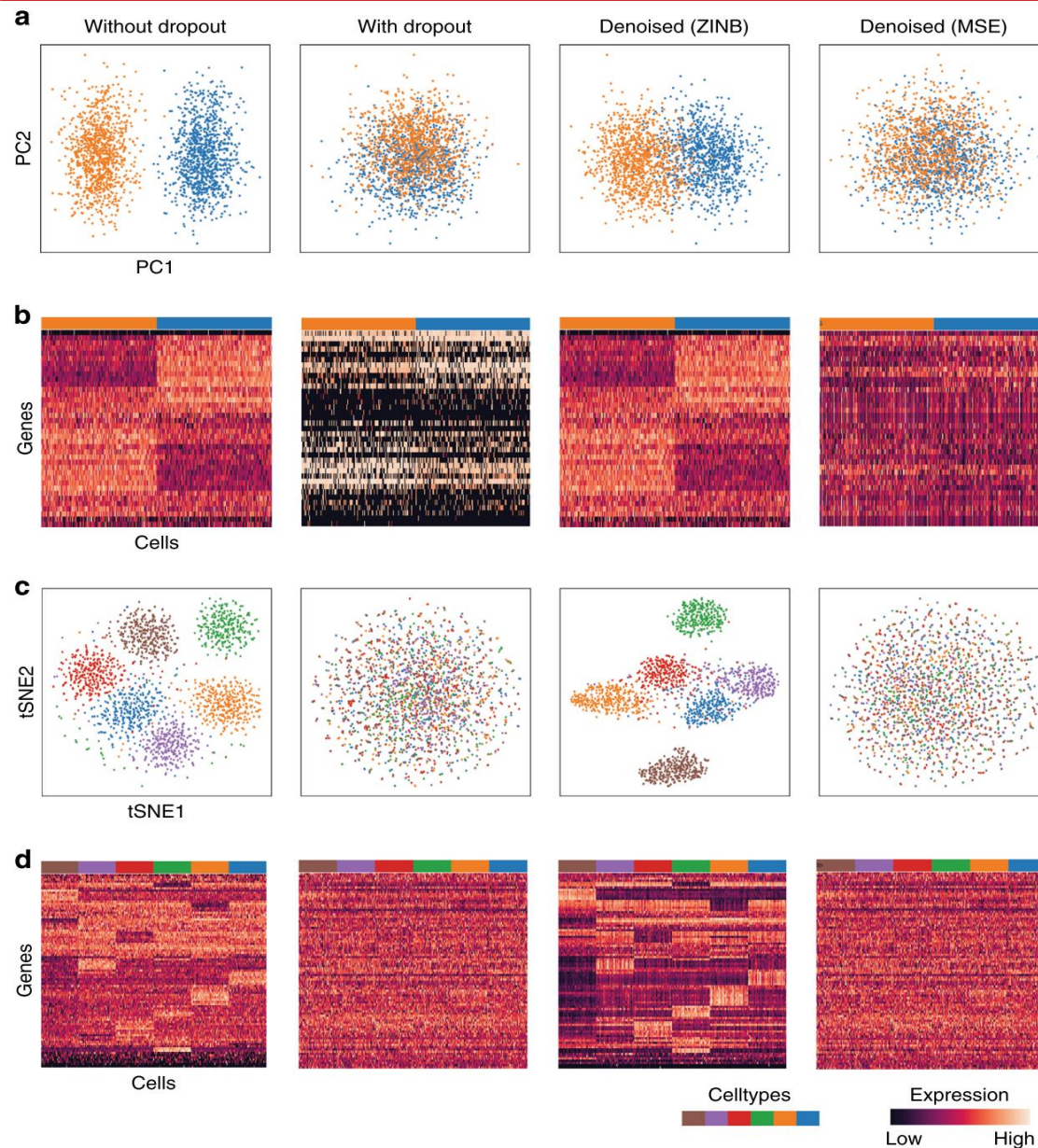


- The geometric illustration of denoising process is adapted from book “Deep Learning” by *Goodfellow et al.* 2016
- **Red arrows:** how a corruption process, i.e. measurement noise including dropout events, moves data points x_j away from the data manifold (**black line**)
- **Green arrows:** The autoencoder is trained to denoise the data by mapping measurement-corrupted data points $\tilde{x}_j$ back onto the data manifold



Eraslan et al., Nature Communications, 2019

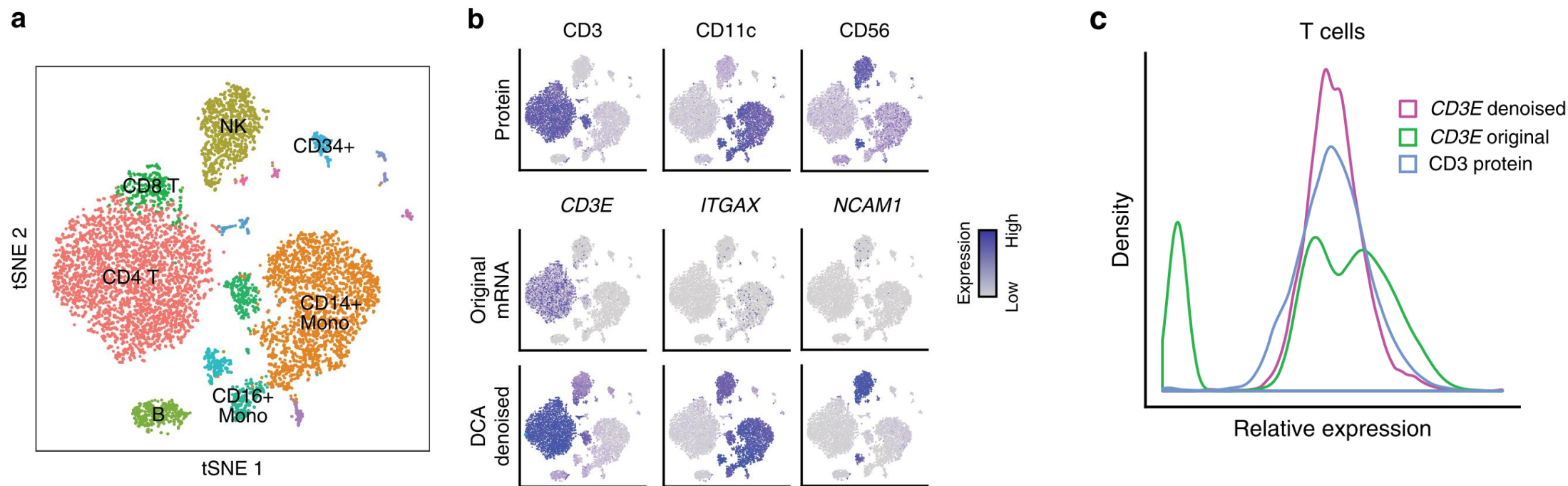
Count-based loss function is necessary



- Is a count-based loss function necessary to identify cell types in simulated data with high levels of dropout noise?
 - The Mean squared error-based autoencoder was unable to recover the cell types
 - The authors claimed that **the specialized count loss function is necessary for the identification of cell types from scRNA-seq data with high dropout noise levels**
- Tip: We should adapt a typical loss function to appropriate noise models for scRNA-seq data

A benefit of using DCA

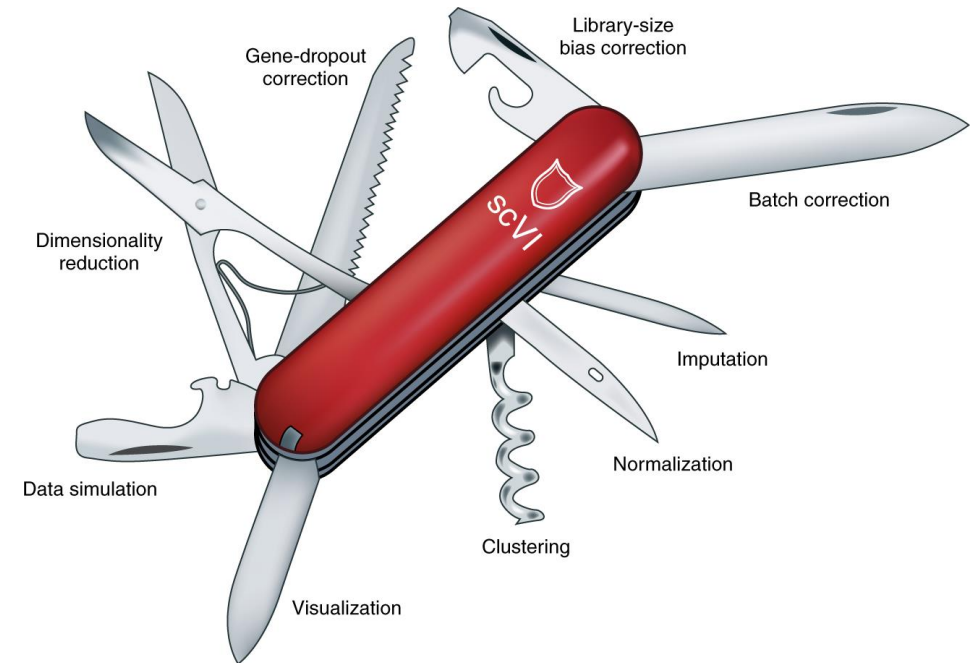
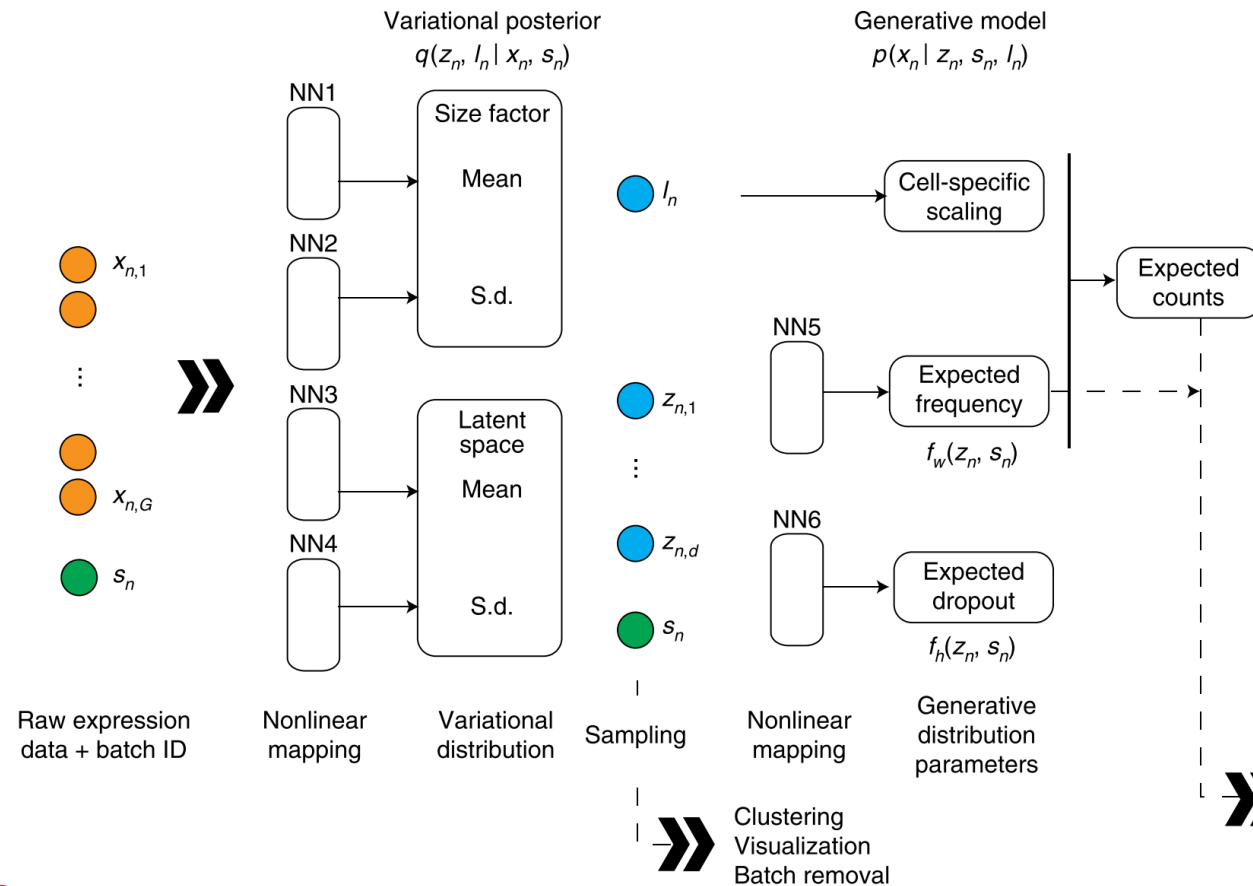
- Autoencoder denoising can increase protein and RNA co-expression



Eraslan et al., Nature Communications, 2019

VAE (variational autoencoder)

- **scVI** is a flexible framework that can be used for multiple tasks (e.g. **batch correction**, visualization, clustering, and differential expression)
 - A probabilistic approach for normalization and multi-task analysis
 - Posterior estimates for distributional parameters
 - Explicitly model library size and batch effects

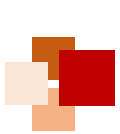


Lopez et al., Nature Methods, 2018
Way and Greene, Nature Methods, 2018



Spatial Transcriptomics





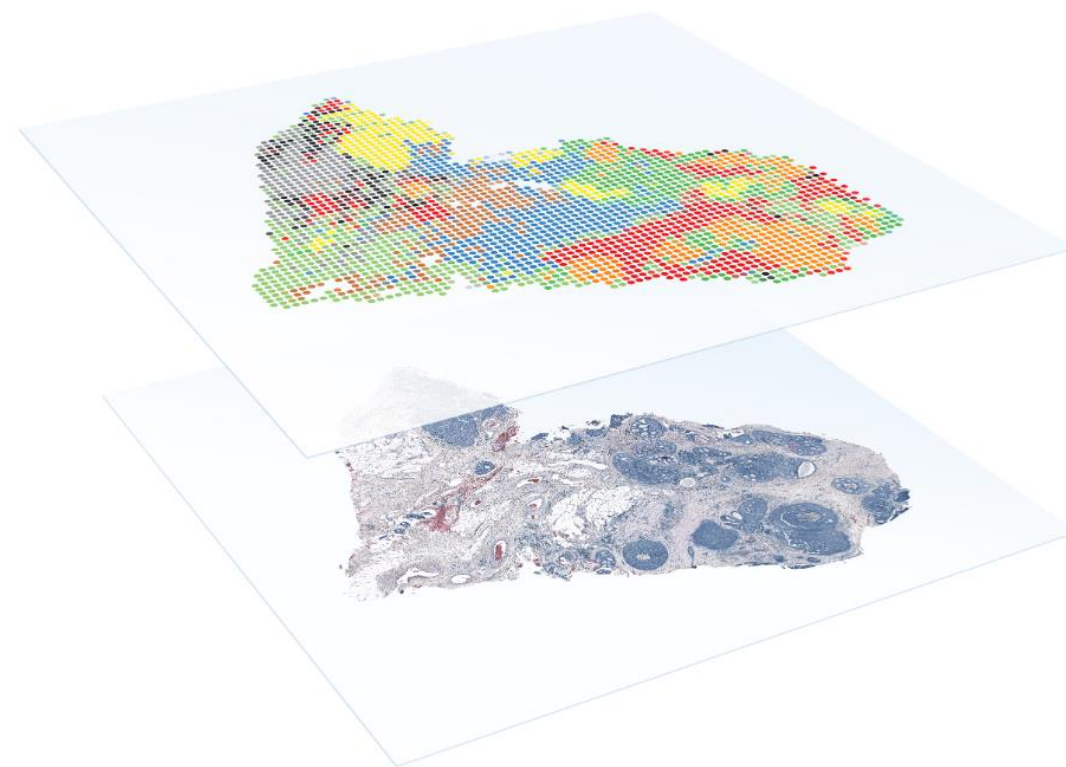
What is spatial transcriptomics?

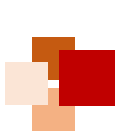


Spatial transcriptomics attempts to quantify mRNA expression of large numbers of genes within the spatial context of tissues and cells



Moses et al, Nature Methods, 2022



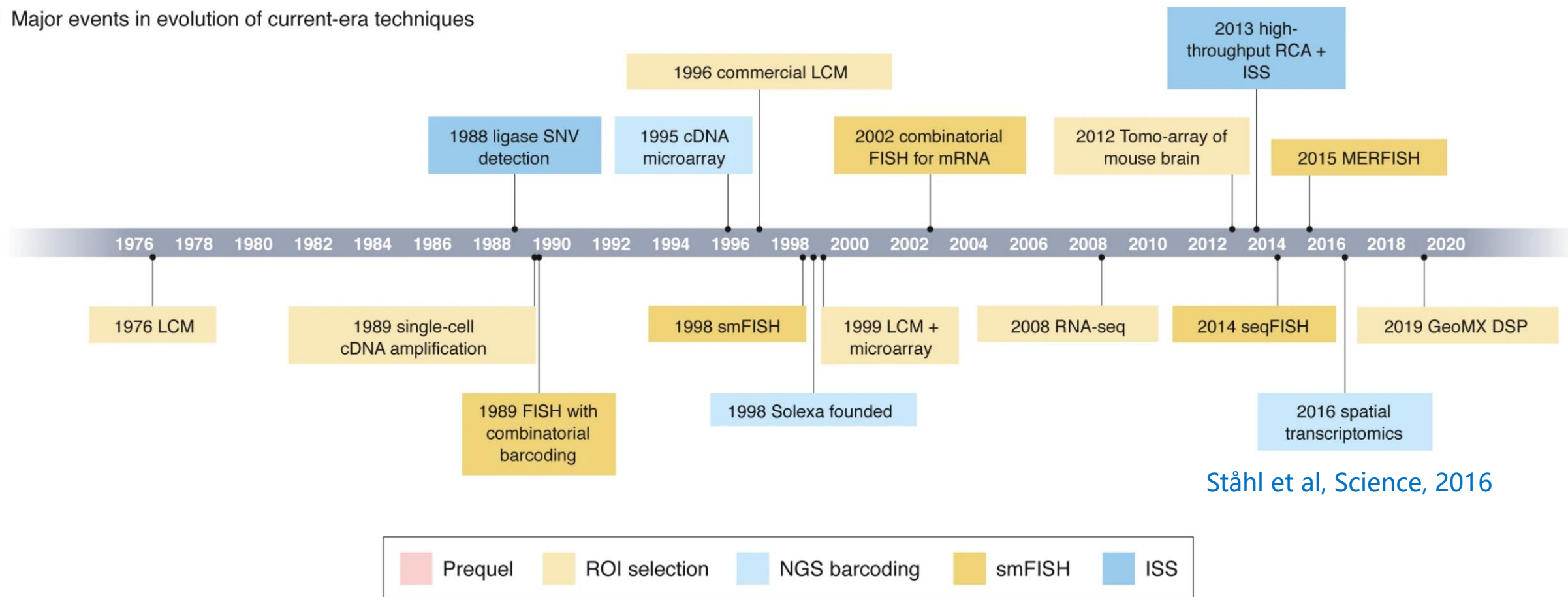


Brief history of spatial transcriptomics



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c Major events in evolution of current-era techniques

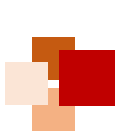


Ståhl et al, Science, 2016

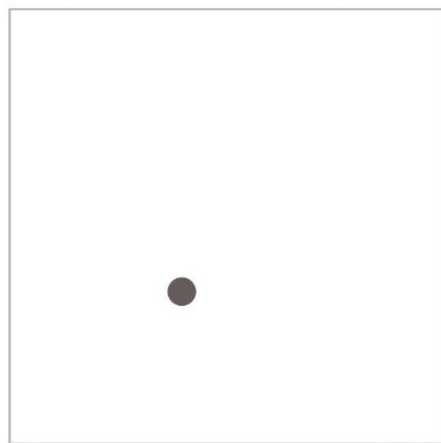
Moses, L. and Pachter, L. Nature Methods, Vol. 19, pp. 534-546, 2022.

Moses et al, Nature Methods, 2022

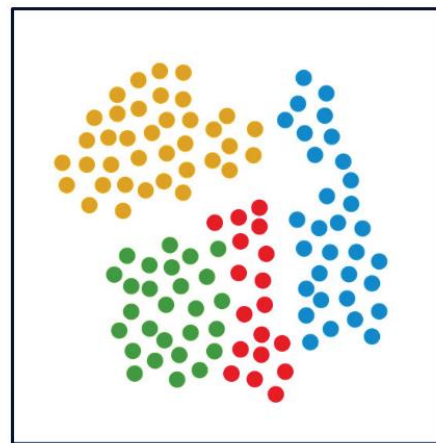
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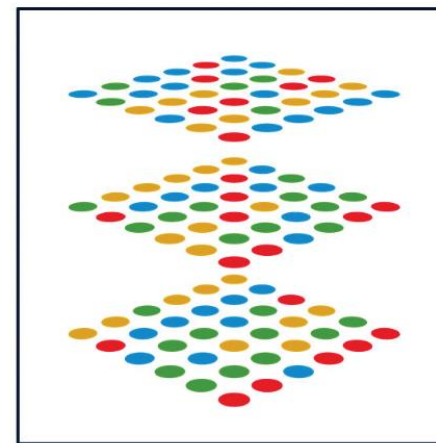
Three RNA sequencing techniques



Bulk



Single Cell

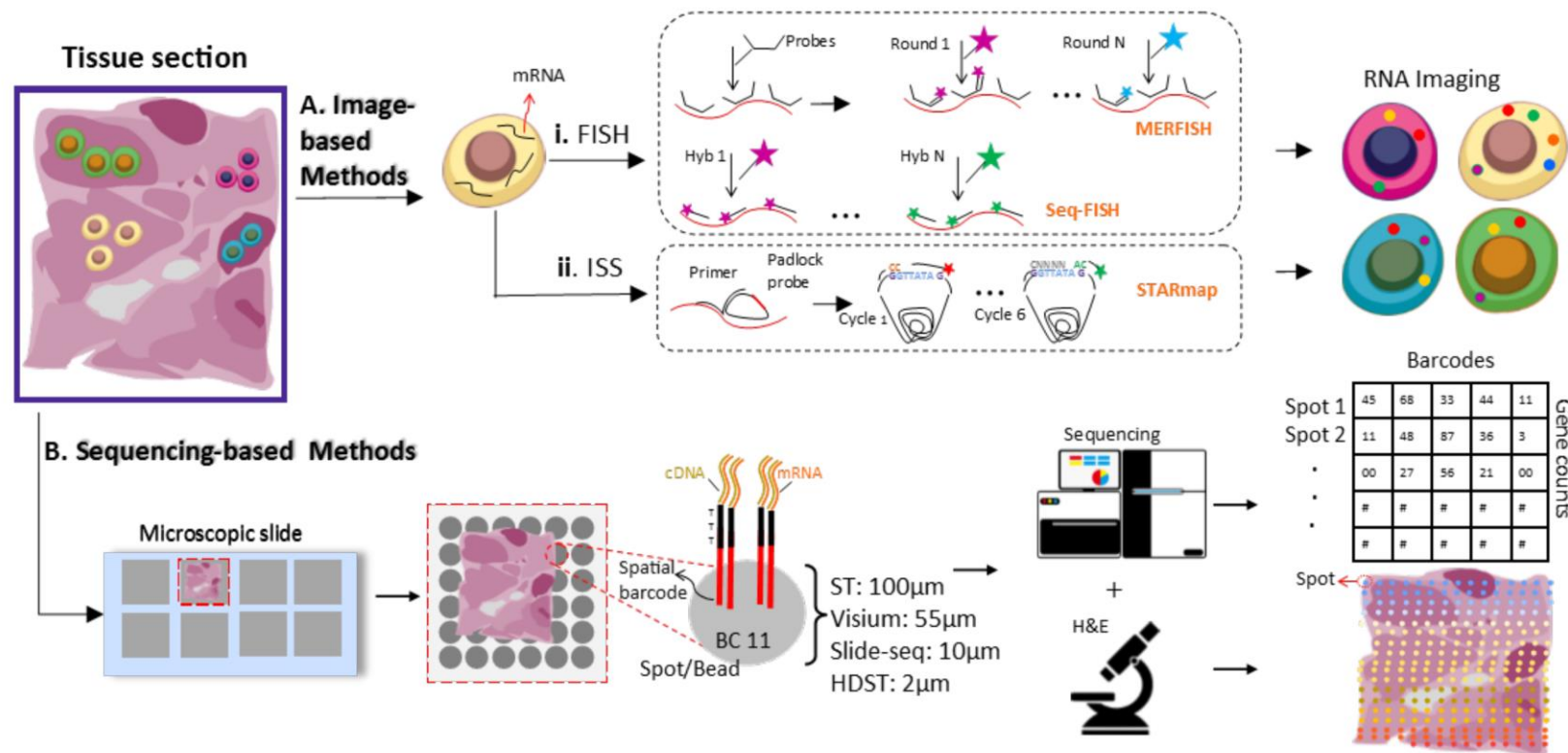


Spatial

- **Bulk RNA-seq**: measures average gene expression from the mRNAs in a tissue' s cells — its transcriptome
- **Single-cell RNA-seq**: enables access to the transcriptomes of thousands of single cells
- **Spatial transcriptomics**: not only provide transcriptomics data, but also retain the positional context of cells in a tissue



Two types of spatial transcriptomics methods



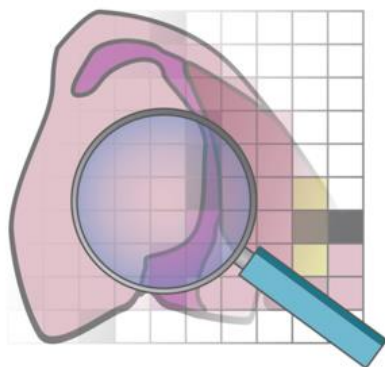
- **Image-based methods**: profile a few genes at high spatial resolution (single molecular level) and high sensitivity. e.g. MERSCOPE (based on MERFISH, <100nm, hundreds genes), Xenium (10x Genomics)
- **Sequencing-based methods**: profile many genes with limited spatial resolution (>10μm), e.g. Visium (10x Genomics, 55μm, 18000 genes), GeoMX (NanoString)

Applications of spatial transcriptomics

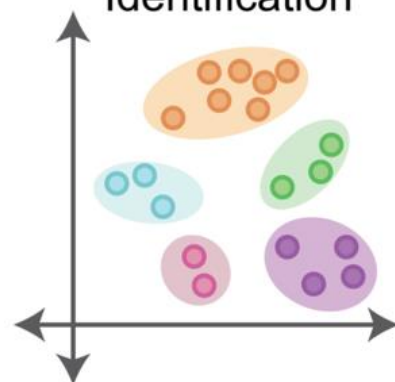


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A Image Analysis



B Cell Type Identification



C Deconvolution



D Spatial Distribution



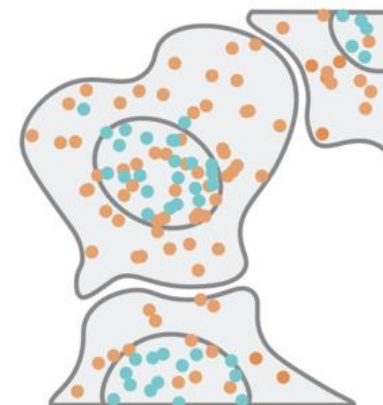
E Cell-Cell Interaction



F Spatial Expression Patterns



G Subcellular



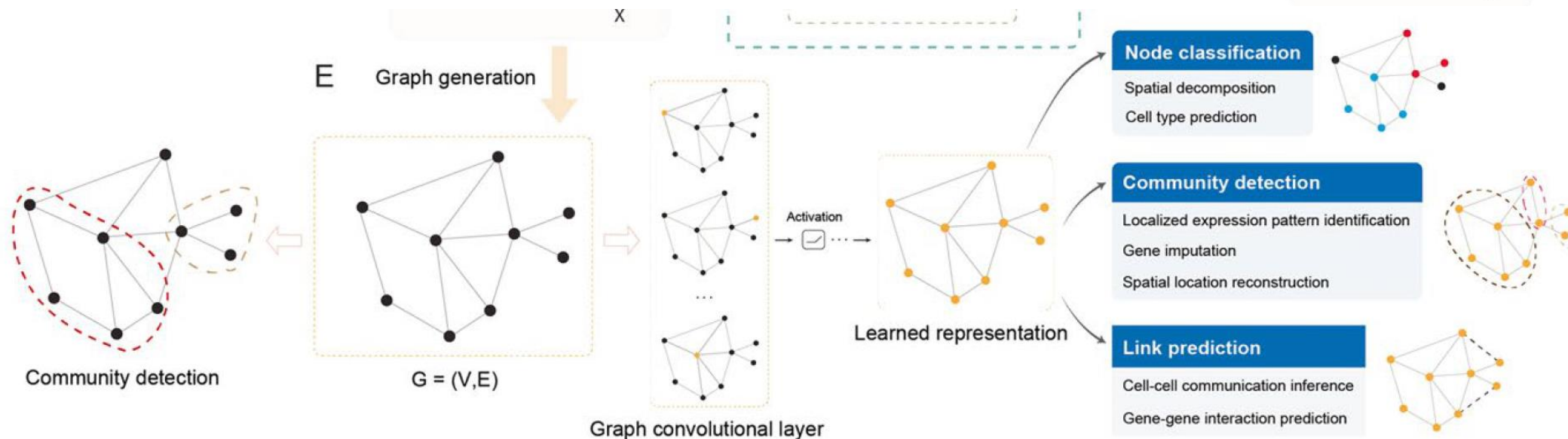
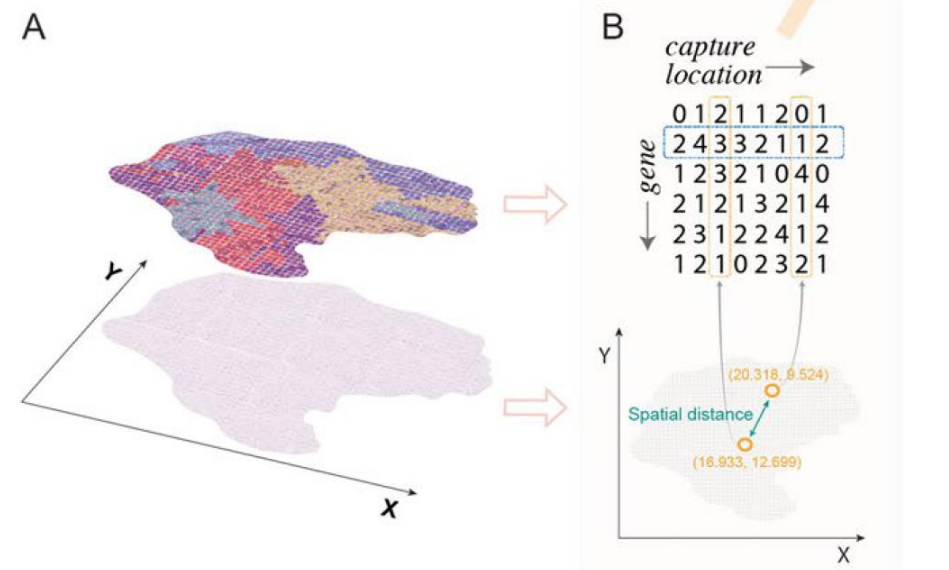
Dries et al. Genome Research, 2021

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Cell type identification



- **Problem:** Spatial Clustering
- **Graph modeling:** Each node corresponds to a spatial location, and two nodes are connected if they have proximate locations or similar expression profiles





Recap



- In this lecture, we have learned:
 - The meaning and measurement of gene expression
 - How RNA-seq data are generated
 - A typical workflow of RNA-seq data analysis
 - The motivation for single-cell RNA-seq (scRNA-seq) in regard to limitations of bulk RNA-seq
 - Computational challenges in the processing of scRNA-seq data
 - Autoencoder is a deep learning method widely used for scRNA-seq data processing and analysis
 - The techniques and applications of spatial transcriptomics





References



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